**Abstract**

This study focuses on the development of the JHCSC Interactive Information Access, a web-based system that utilizes QR code technology to improve access to information about trees and school buildings within the J.H. Cerilles State College (JHCSC) main campus. With the rapid advancement of technology, QR codes have become an effective tool for providing quick and efficient access to digital information. However, many institutions still rely on traditional methods such as printed labels and signboards, which are often outdated, difficult to maintain, and limited in the amount of information they can provide. The proposed system addresses these challenges by assigning unique QR codes to each tree and building on campus. When scanned using a smartphone or tablet, users can instantly access detailed and updated information. For trees, this includes the common name, scientific name, family, ecological and conservation status, and maintenance history. For buildings, relevant structural and administrative information is provided. The system is supported by a centralized database and managed by authorized personnel who ensure that all data remains accurate and up to date. The development of the system follows a structured methodology, incorporating data gathering, system design, development, testing, and implementation. By integrating digital technology with campus resource management, the system enhances learning, supports research activities, and promotes environmental awareness and institutional transparency. Furthermore, it provides an efficient platform for stakeholders, including students, faculty, visitors, and researchers, to access reliable information in real time. The JHCSC Interactive Information Access demonstrates how QR code technology can be effectively applied in educational institutions to modernize information systems, improve accessibility, and contribute to sustainable campus management practices.

*Keywords: QR Code Technology; Web-Based System; Information Access; Campus Resource Management; Environmental Awareness*

**JH CERILLES STATE COLLEGE**
**School of Computing Studies**
*Mati, San Miguel, Zamboanga del Sur*

# JHCSC INTERACTIVE INFORMATION ACCESS

A CAPSTONE PROJECT
Presented to the faculty of
School of Engineering and Technology

In Partial Fulfilment
Of the Requirements for the Degree

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

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2026

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Researchers

# EXECUTIVE SUMMARY

The advancement of information technology has transformed the way educational institutions manage and disseminate information. In higher education institutions, the use of web-based platforms and QR code technology improves transparency, accessibility, and sustainability. At J.H. Cerilles State College (JHCSC) Main Campus, information about campus trees and school buildings is traditionally displayed through physical boards and printed labels. However, these methods are often incomplete, difficult to update, and not easily accessible to students, faculty, visitors, and researchers.

To address these challenges, the JHCSC Interactive Information Access system was developed. This web-based platform integrates QR code technology with a centralized database to digitize and manage information about both trees and school buildings within the campus. Each tree and building is assigned a unique QR code that can be scanned using a smartphone or tablet. Once scanned, users can instantly view detailed profiles.

The system was developed using the 4D Model, comprising the Discover, Design, Develop, and Deliver phases, ensuring a systematic approach. Through data gathering, system analysis, interface design, database development, testing, and deployment. By integrating QR code technology and modern web development tools, the JHCSC Interactive Information Access system improves efficiency in accessing campus information, enhances environmental awareness, supports research and learning activities, and promotes better management of institutional and natural resources.

# TABLE OF CONTENTS


| Content | Page |
|---|---|
| APPROVAL SHEET | ii |
| ACKNOWLEDGEMENT | iii |
| EXECUTIVE SUMMARY | iv |
| TABLE OF CONTENTS | v |
| **CHAPTER 1- PROBLEM AND ITS SCOPE** | |
| Introduction | 1 |
| Project Context | 2 |
| Purpose and Description | 4 |
| Objectives | 5 |
| Scope and Limitation | 5 |
| Definition of Terms | 6 |
| **CHAPTER II – REVIEW OF RELATED LITERATURE** | |
| Related Studies | |
| Technical Background | 8 |
| Hardware Specifications | 9 |
| Software Specifications | 10 |
| Programming Environment | 10 |

Foreign 11

Local 15

**CHAPTER III – DESIGN AND METHODOLOGY**

Requirements Analysis 18

Discover Phase 19

**CHAPTER IV – DEVELOPMENT, TESTING, AND IMPLEMENTATION**

Description and Prototype 32

Development Phase 34

Store Data Module 34

Generate QR Code Module 36

Scan QR Code Module 36

Testing

Alpha Testing 39

Beta Testing 44

Implementation Plan 45

**CHAPTER V – IMPLEMENTATION RESULT, DISCUSSION, CONCLUSION, RECOMMENDATION**

Implementation Result and Discussions 46

Results and Discussions 49

Conclusion 51

Recommendation 51

**REFERENCES** 52

**APPENDICES**

A. Test Cases 55

B. Sample Output 61

C. Photo Documentation 62

D. Curriculum Vitae 63

# LIST OF FIGURES

**Figures** **Page**

1) 4D Model 18

2) Current Technical Situation 20

3) Context Diagram 21

4) Conceptual Framework – IPO 22

5) Activity Diagram (Administrator) 23

6) Activity Diagram (Instructor/Administrative Personnel) 24

7) Activity Diagram (Public Users) 25

8) Use Case Diagram 26

9) Entity Relationship Diagram 28

10) Structural Diagram (Administrator) 29

11) Structural Diagram for Instructor/Administrative Personnel 30

12) Structural Diagram (Public Users) 30

13) System Architecture 31

# LIST OF TABLES

**Tables** **Page**

1) Use Case Description for Generate QR Code 27

2) Use Case Description for Scan QR Code 27

3) Gantt Chart 32

4.0) Implementation Plan 45

# LIST OF IMAGES

**Images** **Page**

1) Data Gathering 20

2) Beta Testing 44

3) Manage Tree Data 46

4) Add Tree Data 47

5) View Tree Data 47

6) Add Building Data 48

7) View Building Data 48

8) View Scanned QR Code 49

# CHAPTER I

# BACKGROUND OF THE STUDY

## Introduction

Today, technology is advancing rapidly and changing how people live, work, and learn. One of these technologies is the Quick Response (QR) code, which is now widely used because it gives users quick access to information by simply scanning it with their phones (Arakelyan, 2017). QR codes are not only used in businesses but are now also applied in schools, hospitals, government offices, and even in environmental protection. During the COVID-19 pandemic, the Department of Trade and Industry (DTI) in the Philippines encouraged the use of QR codes for cashless payments to keep people safe and help businesses operate more smoothly (Department of Trade and Industry, 2020). This shows how QR codes can be applied in many areas to make sharing and accessing information faster and easier.

Even though sharing information is now easier, many schools still struggle to provide up-to-date details about important resources and facilities on their campuses, including trees and even school buildings. For example, schools with large campuses often have a wide variety of trees that are useful for learning and research, but this information is hard to find. Similarly, school buildings, such as classrooms, laboratories, libraries, and offices, contain details. However, these details are usually managed through outdated methods like paper records, signboards, or name tags that fade or become outdated. These traditional methods are not always accurate or easy to update, and they do not fully help students, teachers, or visitors understand and appreciate the resources and facilities available on campus (Sethy & Dadhich, 2023).

Not having easy access to updated information about both trees and buildings affects a wide range of stakeholders. Students in environmental science and forestry may struggle to access data for their schoolwork or research. Visitors and community members also miss out on opportunities to learn more about the environment and the institution's facilities. According to the Department of Environment and Natural Resources (DENR), public involvement is key to promoting environmental and institutional awareness (Department of Environment and Natural Resources, 2022). Thus, a system that provides quick, reliable, and easily accessible information through QR codes can help address these gaps.

To respond to these concerns, the JHCSC Interactive Information Access will be developed. It will use QR codes to create a simple platform that stores and displays information about both trees and school buildings within the JH Cerilles State College campus. By scanning a QR code attached to a tree, users can view details such as its local name, scientific name, family, ecological status, conservation status, and maintenance history. Similarly, QR codes placed on school buildings will allow users to access information. This study aims to enhance learning, support research, and increase awareness by making environmental and institutional information more accessible, thereby encouraging greater participation in protecting and appreciating the campus’s natural and structural resources.

**Project Context**

At JH Cerilles State College (JHCSC), both natural and structural resources play an important role in the campus environment. Trees contribute to cleaner air, provide shade, and create a peaceful, green atmosphere, while school buildings, such as classrooms, laboratories, libraries, and offices, serve as spaces for learning, research,

and administration. However, one challenge is that visitors, students, teachers, and researchers do not always have updated access to important details about these resources. For example, tree profiles, such as common/local names, scientific names, ecological status, or maintenance history, are often incomplete or outdated. Similarly, school buildings may lack accessible records. While there are some posted boards or markers, not all trees and buildings have them, and much of the data is not regularly updated. This makes it difficult for stakeholders who need accurate information for schoolwork, research, or environmental and institutional activities. Often, they either fail to get the right data or spend extra time asking different people, which reduces efficiency and creates barriers to learning and awareness.

To address these concerns, the JHCSC Interactive Information Access will be developed. This system assigns each tree and building on campus a unique QR code. When scanned with a phone or tablet, the QR code displays details: for trees, common/local name, scientific name, family name, ecological and conservation status, and maintenance history; for buildings, the building name and the offices in it. The information will be recorded, updated, and managed by designated staff and faculty, including instructors from the Bachelor of Science in Forestry (BSF) program in trees and administrative personnel for school facilities. This will benefit not only the school community but also visitors, researchers, and environmental or educational groups, who can use the data for projects, reports, research, and institutional planning.

At a time when the environment and education systems face many challenges, it is important to use technology to provide innovative solutions. Since most students and visitors already rely on smartphones and internet access, a QR code system is a practical and effective way to promote environmental awareness, enhance institutional transparency, and support learning. By adopting this system, JHCSC can serve as a

model for other campuses that aim to integrate green practices with digital innovation. The system will not only improve the management of information about trees and school facilities but will also demonstrate how technology can enrich education, research, and sustainability initiatives.

**Purpose and Description**

The main purpose of this study is to provide a solution to the challenges encountered in accessing information about both trees and school buildings within the J.H. Cerilles State College (JHCSC) main campus using QR code technology. By integrating QR codes, this study aims to make campus data easily accessible anytime, including data related to natural resources such as trees and institutional facilities. It also seeks to encourage the school community to actively protect the environment, appreciate campus facilities, and support sustainable practices.

The proposed JHCSC Interactive Information Access system is designed to record, organize, and display updated information about the college's trees and buildings. Each tree and building will be assigned a unique QR code that can be scanned using a phone or tablet. When scanned, the QR code for a tree will display details such as its common/local name, scientific name, family name, ecological status, conservation status, and maintenance history. For school buildings, the QR code will provide information about the building. This system will not only support learning and research but also promote environmental awareness, institutional transparency, and improved campus management, making JHCSC a model for integrating technology with education and sustainability.

## General Objectives

This study aims to develop the JHCSC Interactive Information Access, a web-based system designed to manage and provide easy, real-time access to detailed information about both trees and school buildings within the JHCSC Main Campus using QR code technology.

## Specific Objectives

The specific objectives are:

- To create a digital system to manage the profiles of both trees and school buildings on the campus.
- To design a user-friendly QR code system that can be easily scanned by public users using mobile devices to view the profiles of trees and buildings in real time.
- To develop a web-based platform where tree data can be viewed and updated in real time.
- To implement the system across the campus by placing QR codes on trees and school buildings to ensure that users can access and interact with the information effectively in the JHCSC Main Campus.

## Scope and Limitations

The JHCSC Interactive Information Access is a web-based platform designed to manage and provide up-to-date information on trees and school buildings on the JHCSC main campus. The system generates a unique QR code for each tree and building, which can be scanned by public users using a smartphone or tablet. Once scanned, the code displays a digital profile of the trees and school buildings. Authorized

users, instructors, and administrative personnel can update the information through the system to ensure that all data remains accurate and up to date. This study aims to support environmental awareness, institutional transparency, research, and proper management of both natural and structural resources within the school.

The system, however, has its limitations. It is currently available only on the JHCSC campus and cannot yet be extended to other locations. For trees, the system does not include automatic sensors to monitor their health, so updates must be done manually. For both trees and buildings, QR codes may get damaged, lost, or faded over time, which could affect accessibility if not properly maintained. Despite these limitations, the system provides a practical step toward digital innovation and sustainability in the campus setting.

**Definition of Terms**

*Administrator* – The admin is responsible for managing all users, approving updates, and overseeing the overall system activities.

*Instructors or Administrative Personnel* – These are registered users who are allowed to update tree and building information in the system.

*Public Users* – These are the users who can scan the QR codes placed on trees and school buildings. Public users can view the tree and building information, but cannot make any changes to the data.

*JHCSC Interactive Information Access* – A web-based system developed to monitor and manage the trees and buildings within the J.H. Cerilles State College (JHCSC) main campus. The system uses QR code technology to provide easy access to information about each tree and building.

*QR (Quick Response) Code* - A type of code that can be scanned using a smartphone to quickly access information, specifically used in this study to link to a tree's and building's digital profile.

*Tracking* – This refers to the act or process of managing and keeping records of individual trees and school buildings on the campus.

*QR Code Generator* – This system feature automatically generates a unique QR code for every tree added to the database. Each generated code links directly to a specific tree profile in the system.

*Real-Time Update* – The process of real-time update means that when a forestry instructor updates a tree's profile, the changes are reflected immediately after saving. So, when the user scans the QR code, they see the most recent information.

# CHAPTER II

# REVIEW OF RELATED STUDIES AND SYSTEMS

This chapter covers the technical background, including hardware and software specifications, the programming environment, and foreign and local studies.

## Technical Background

The JHCSC Interactive Information Access is a web-based platform designed to support environmental awareness, institutional transparency, and promote fast access to both tree and building information on the J.H. Cerilles State College (JHCSC) main campus. This system aims to address the current challenges in accessing and updating campus information by assigning each tree and school building a unique QR code linked to a digital profile. Through this system, public users can easily access details of a tree's profile or a building's information, making the campus more organized, informative, and sustainable (Sethy & Dadhich, 2023).

QR codes are 2D barcodes that store information and can be scanned with a device's camera (Arakelyan, 2017). These are extremely readable and can store large amounts of data for various applications. In the JHCSC Interactive Information Access, QR codes will be created and assigned to every tree and school building as their digital tags, containing the profile of each tree and building. The information obtained using QR codes is accessible, more secure, and easier to update.

Web-based systems are accessible via web browsers such as Google Chrome, Mozilla Firefox, or Safari. The functionality of web systems is to be used over the internet, enabling their users to view, retrieve, and manage information without being physically confined to a specific location or installing software (Laudon & Laudon, 2020). It is very accessible and easy to use. JHCSC Interactive Information Access can

be accessed by public users with a stable internet connection to view information about the trees. Having web-based functionality ensures that tree and building information is easily accessible, always up to date, and under central management.

The database provides a structured way to store and manage information about both trees and school buildings within the JHCSC campus. This centralized storage enables users and systems to easily retrieve, filter, and update data quickly and efficiently. By integrating both natural and structural resources, the system ensures data consistency, reliability, and accessibility for learning, research, and campus management (Ramakrishnan & Gehrke, 2018).

**Hardware Specifications**

| **Hardware** | **Description** |
|---|---|
| Processor | Intel Core i3 |
| RAM | 4GB |
| Hard Disk Drive | 300GB |
| Camera | at least 8MP, capable of QR code scanning |
| Printer | Inkjet Printer |

**Software Specifications**

| **Software** | **Description** |
|---|---|
| **Server Environment** | |
| Operating System | Windows 10 |

| | |
|---|---|
| Browser | Google Chrome, Brave Browser |
| XAMPP | Version 2.1.1 (includes Apache, MySQL, PHP) |
| Visual Studio Code | Version 1.84 (code editor for PHP and web development) |
| **Client Environment** | |
| Web Hosting Server | Infinity Free (1GB storage, for hosting live system) |

**Programming Environment**

HTML (HyperText Markup Language) is used to structure the content and layout of the web pages. It serves as the backbone of the system's frontend by defining the placement of text, images, input fields, buttons, and other elements. With HTML, the system can clearly and efficiently present tree profiles and user options to all types of users.

CSS (Cascading Style Sheets) is applied to HTML to style the interface and enhance the visual presentation of the system. CSS allows customization of fonts, colors, spacing, alignment, and responsiveness, ensuring that the system is not only functional but also aesthetically pleasing and easy to navigate, whether accessed through desktops, laptops, or mobile devices.

PHP (Hypertext Preprocessor) is used as the main server-side scripting language. It handles the system's dynamic processes, including managing the database, processing user input, retrieving tree information, and generating QR codes. PHP enables the system to perform operations like saving new tree profiles, updating existing data, and controlling user access levels.

XAMPP is the local development environment used to host the system during development and testing. It is a free and open-source platform that includes Apache (web server), MySQL (database), and PHP (scripting language interpreter). XAMPP allows developers to simulate the server environment on a local computer before deploying the system online.

HTML, CSS, PHP, and XAMPP will be used as the development stack for JHCSC Interactive Information Access. HTML provides the structural elements for the user interface, while CSS enhances its visual appeal and responsiveness. PHP serves as the engine behind the scenes, enabling dynamic data management, QR code generation, and secure database interactions. XAMPP provides a temporary local server environment, allowing convenient development, testing, and debugging before deployment, ensuring a fully functional, user-friendly solution for managing and accessing tree information on the JHCSC campus.

## Related Studies

### Foreign

*QR Code Application in Tree Management: A Pilot Project*

According to Le et al. (2023), this system uses QR code technology to enhance access to information about campus structures, including plants at Can Tho University in Vietnam. The concept involves generating unique QR codes for each item, which, when scanned by a smartphone, provide immediate access to detailed information stored in a centralized database. The system incorporates a user-friendly website interface that allows administrators to efficiently manage and update data, ensuring the information remains current and accurate. Field testing confirmed the application's functionality, demonstrating its effectiveness in providing details about scanned items.

This innovation aims to offer visitors and students a simple, interactive way to learn about their environment.

This study relates to the JHCSC Interactive Information Access since it also adopts the concept of using QR code technology to provide quick and reliable access to information. Similar to the pilot project, which focused on managing and presenting details about trees through QR codes, the JHCSC system applies the same principle but extends it further by including both trees and school buildings within the campus. While the previous study highlighted the role of QR codes in enhancing environmental awareness and interactive learning, the JHCSC Interactive Information Access expands this purpose.

*Identification of Plant Species by Deep Learning and Providing as A Mobile Application*

According to Adak, M. (2020), this study focuses on developing a QR code system specifically designed to identify tree species, enhance public knowledge, and improve tree management practices at Sakarya University, Turkey. The system was developed using a three-tier architectural model, integrating a PHP script for backend operations, a QR code library for generating unique codes, and a website database to store detailed tree information. Once generated, these QR codes are physically attached to trees, allowing users to simply scan them with their mobile devices to instantly launch a website displaying the corresponding tree's data. A survey of end users shows that the system is user-friendly and makes tree identification more engaging and interesting.

This study relates to the JHCSC Interactive Information Access, as it also uses QR code technology to provide accessible information about trees. Similar to

Identification of Plant Species by Deep Learning and Providing as A Mobile Application, which developed a QR-based system for identifying tree species and improving public awareness. The JHCSC system likewise assigns unique QR codes to each tree to store and display important details such as scientific name, family, and ecological status. However, the JHCSC system goes beyond tree identification by also incorporating information about school buildings, thereby expanding its scope to both natural and structural resources. In doing so, it not only supports environmental education and awareness but also enhances institutional transparency and campus resource management.

*Development of a QR Code System for Tree Species Identification*

According to Onyekwelu et al. (2023), this research focused on developing a more efficient and accessible QR code system for tree identification in Ondo, Nigeria. Data for this system were collected from tree populations on the campuses of two tertiary institutions in Akure, Nigeria, and the Federal University of Technology and the Federal College of Agriculture. It has a QR code generator feature developed with PHP 8.1. Each generated QR code is directly linked to a website database containing information on all three species, allowing users to simply scan a code attached to a tree to automatically launch the website and retrieve its details.

This study relates to the JHCSC Interactive Information Access since it also focuses on the use of QR code technology to provide quick access to tree species information. Similar to the research above, which generated QR codes linked to a centralized database for tree identification, the JHCSC system also enables users to scan QR codes attached to trees to retrieve detailed profiles, including common/local name, scientific name, and ecological status. However, while the Nigerian study focused

solely on tree species, the JHCSC system extends its application by including both natural and structural resources, specifically trees and school buildings on campus. This broader scope enhances not only environmental learning and awareness but also institutional transparency and management, making the system more comprehensive in addressing the needs of its stakeholders.

*QR Code Generator Application to Access Information on the Campus Building*

As stated by Wahyuni et al. (2023), this study developed an Android-based QR code generator application designed to provide quick access to information about campus buildings at Universitas Pembangunan Panca Budi, Indonesia. The system integrates QR codes with a real-time database, enabling users to scan codes placed on buildings to instantly retrieve details such as location, facilities, and relevant administrative information. A web interface was also implemented for administrators to easily manage and update building data, ensuring accuracy and consistency. Field testing showed that the application worked effectively, demonstrating the usefulness of QR codes in improving access to building information and enhancing campus services.

This study relates to the JHCSC Interactive Information Access, as it also focuses on using QR code technology to improve the accessibility and management of school infrastructure. While the QR Code Generator Application to Access Information on the Campus Building centered on providing information about campus buildings through QR code integration, the JHCSC system extends this application to both school buildings and natural resources such as trees. Both systems highlight the adaptability of QR codes for digitizing institutional assets, improving transparency, and ensuring that updated information is easily accessible to students, staff, and visitors. This

similarity emphasizes the role of QR code technology in promoting efficiency and sustainability across different educational contexts.

**Local**

*Magpahalaga: Appreciating UP Diliman's History and Culture Through QR Codes*

As stated by Ibañez, M. C., (2024), the University of the Philippines Diliman implemented the "Balik Yapak" and "Magpahalaga" projects to install QR codes on key historic buildings around campus. When scanned, each QR code directs users to webpages with detailed information about the building's name, architect, year of construction, occupants, and history. Signs with QR codes are placed in front of pioneer buildings, such as Melchor Hall and Quezon Hall, to enhance awareness of campus heritage and architecture among visitors, students, and staff. The project also includes multilingual options (English, Filipino) and aims to make the campus experience more interactive and educational.

This study relates to the JHCSC Interactive Information Access since it also uses QR code technology to provide quick access to information about campus buildings. In both cases, mobile device scanners allow users to immediately view profiles or histories of physical infrastructure. Both projects demonstrate the value of QR codes for enhancing transparency, education, and appreciation of campus resources, though JHCSC aims for a broader integration of environmental and institutional information.

*Feasibility in Implementing QR Code Inventory System*

As stated by Camua, J., et al. (2024), this study investigates the feasibility of implementing a QR Code-Based Tool Management System at DPR-Philippine Academy in Technology and Sciences (DPR-PATS). The research identifies current challenges in tool management and proposes a technological solution to enhance efficiency and improve inventory practices. The study highlights the transformative potential of QR code technology to modernize inventory practices and improve academic operations through evaluation. While this research specifically focuses on managing tools within an institutional setting, it offers valuable insights into the effective application of QR code technology for inventory, tracking, and general asset management.

This study relates to the JHCSC Interactive Information Access, as it demonstrates how QR code technology can be applied to improve the management and accessibility of institutional resources. In the research above at DPR-PATS, QR codes were used to modernize tool inventory practices, improving tracking and oversight. In the same way, the JHCSC system uses QR codes to organize and provide easy access to updated information, but instead of tools, it focuses on both natural resources (trees) and structural assets (school buildings). This parallel shows the adaptability of QR code technology across different institutional settings, proving that it can serve as an effective solution not only for inventory management but also for enhancing environmental awareness, transparency, and campus sustainability.

*Development of Records Tracking Management System with QR Code*

As stated by Piloton et al. (2023), the Records Tracking Management System incorporates QR code technology to ease the management and tracking of documents.

Functionalities include the ability for authorized users to upload scanned documents, add new trail entries, view trail details, print QR details, track documents, and efficiently search documents using QR codes. The system's design caters to the needs of different users, including secretaries, staff, and faculty, enabling them to quickly find information or records associated with a scanned code.

This study relates to the JHCSC Interactive Information Access, as it also uses QR code technology to simplify tracking and information retrieval. As highlighted by the study above, the Records Tracking Management System enabled authorized users to manage documents more efficiently by uploading, searching for, and retrieving records linked via QR codes. Similarly, the JHCSC system applies the same principle of quick access and efficient data management, but instead of documents, it focuses on trees and school buildings within the campus. By adapting the concept from document tracking to environmental and institutional resource management, the JHCSC system demonstrates the versatility of QR code technology in improving accessibility, organization, and transparency across different contexts.

The data from foreign and local studies shown above provides a foundational and contextual guidance for the development of the JHCSC Interactive Information Access. The studies, focused on QR code-based identification and management systems, provide direct insights into technological implementation, system architecture, and the benefits of digitizing information for educational and environmental purposes.

# CHAPTER III

# DESIGN AND METHODOLOGY

This chapter presents the methods used in developing the JHCSC Interactive Information Access. It explains the diagrams that illustrate the system's architecture and primary functions. To enhance understanding, each diagram has a detailed explanation, providing insight into the design logic

## Requirements Analysis

The requirement analysis includes the visualization of operations and system flow to determine how the system processes data and its overall activity and structures. Moreover, the methodology used in this study is the 4D Model. This model is a systematic instructional design framework, is particularly suitable for developing learning materials and systems (Utaminingsih et al., 2025).




**Figure 1.** 4D Model

Figure 1 above illustrates the methodology used by the proponents. It is the 4D model, which includes the phases: Discover, Design, Develop, and Deliver. These phases are essential to the study as they serve as the foundation for the system's development. The model guides the proponents on the actions they need to take in order to fully develop the system. The 4D model is considered effective in software development because it provides a structured framework that supports clear goal-setting, efficient planning, and continuous improvement throughout the development

lifecycle (Podeswa, 2014). Compared to traditional linear models such as the Waterfall model, the 4D model offers greater adaptability to changing user requirements and supports more collaborative design and testing, which are critical for delivering functional, user-centered software systems (Pressman & Maxim, 2020).

**Discover Phase**

This phase is the first stage of the 4D model, the Discover phase, which enables the proponents to conduct interviews and gather data. In this stage, the proponents conducted a verbal interview with the forestry instructor at JHCSC using a guide. The interview focused on how the institution manages tree management and building information, how the forestry department oversees this operation, and the challenges they currently face. The data gathered provided insights into existing processes and issues in tree and building information management. After collecting the information, the proponents analyzed it to identify the core problems and determine possible solutions. This phase serves as the starting point of the development process, guiding the proponents with a clear understanding of the situation and laying the foundation for the next stages of the 4D model.




Image 1A. Data Gathering Part 1

Image 1A above illustrates the documentation of the data gathering conducted by the proponents. They interviewed Ms. Ivy M. Molina, a forestry instructor. Ms. Ivy

explained the current procedures for updating the information on the board. This serves as the foundation for the proponents as they proceed to the next stages of the study.



Image 1B. Data Gathering Part 2

Image 1A, shown above, illustrates part 2 of the data gathering conducted by the proponents regarding the school building information. On the left side of the image, the proponents interviewed Mrs. Desiree M. Luste, the Records Officer of J.H. Cerilles State College, to obtain information about the school's buildings. On the right side, the proponents interviewed Mr. Dante Bayocot to collect information about the buildings of J.H. Cerilles State College.

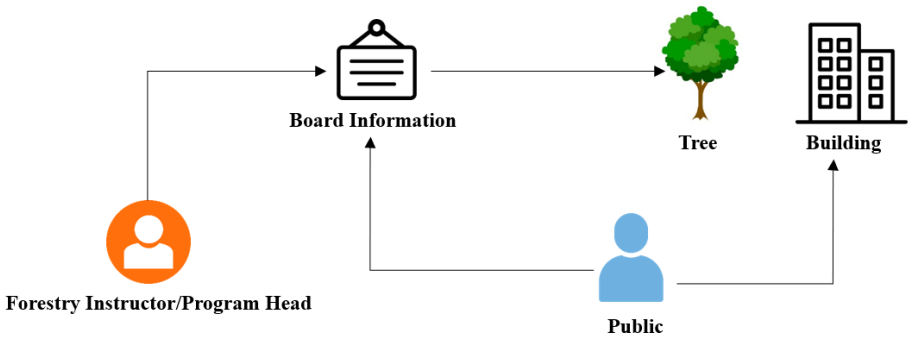



**Figure 2.** Current Technical Situation

The diagram shown in Figure 2.0 illustrates the current process by which different entities provide and access information about trees and buildings. The forestry instructor or program head writes the information on a board, which is then attached to

the tree and building. When public users want to access the information, they have to go to the tree and building information and read it directly from the board. Also, the public can directly view the building or enter it.

**Design Phase**

In the design phase, the proponent lays out system diagrams to provide an overview of the system's functionalities. The diagrams include the context diagram, activity IPO diagram, activity diagram, use case diagram, entity-relationship diagram, structural diagram, and system architecture diagram.

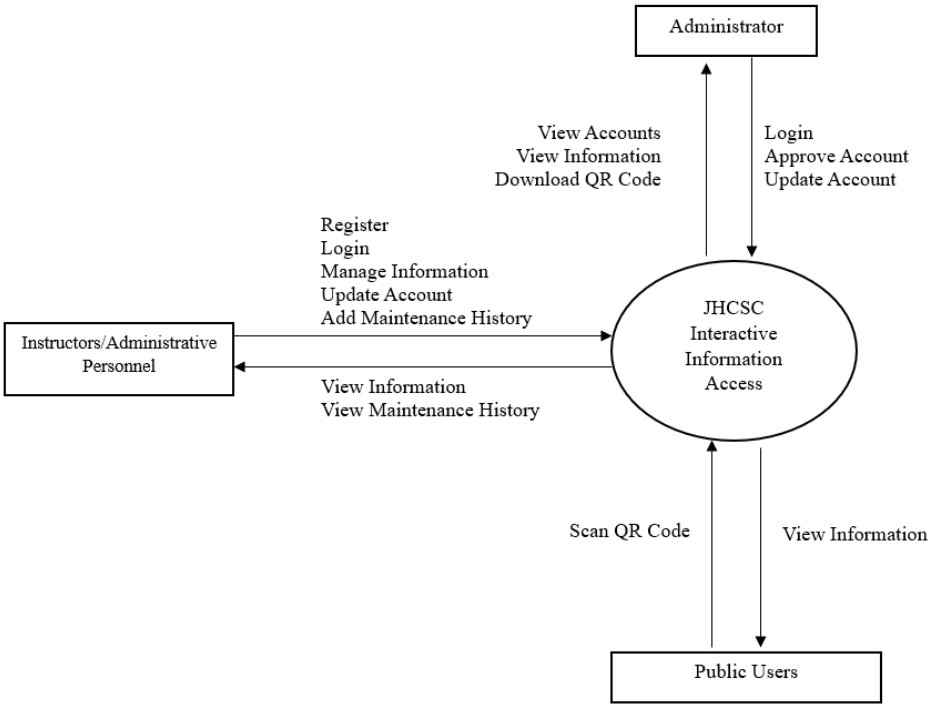



**Figure 3.** Context Diagram

Figure 3 illustrates how users interact with the system and how the system responds to these interactions. The administrator can log in, approve accounts, and update their own account. The system allows the administrator to view accounts, access information, and download QR codes. Instructors or administrative personnel can

register, log in, update their information, add maintenance history, and edit their accounts. In response, the system enables instructors to view information. Public users can access information by scanning a QR code, and the system displays the corresponding information.

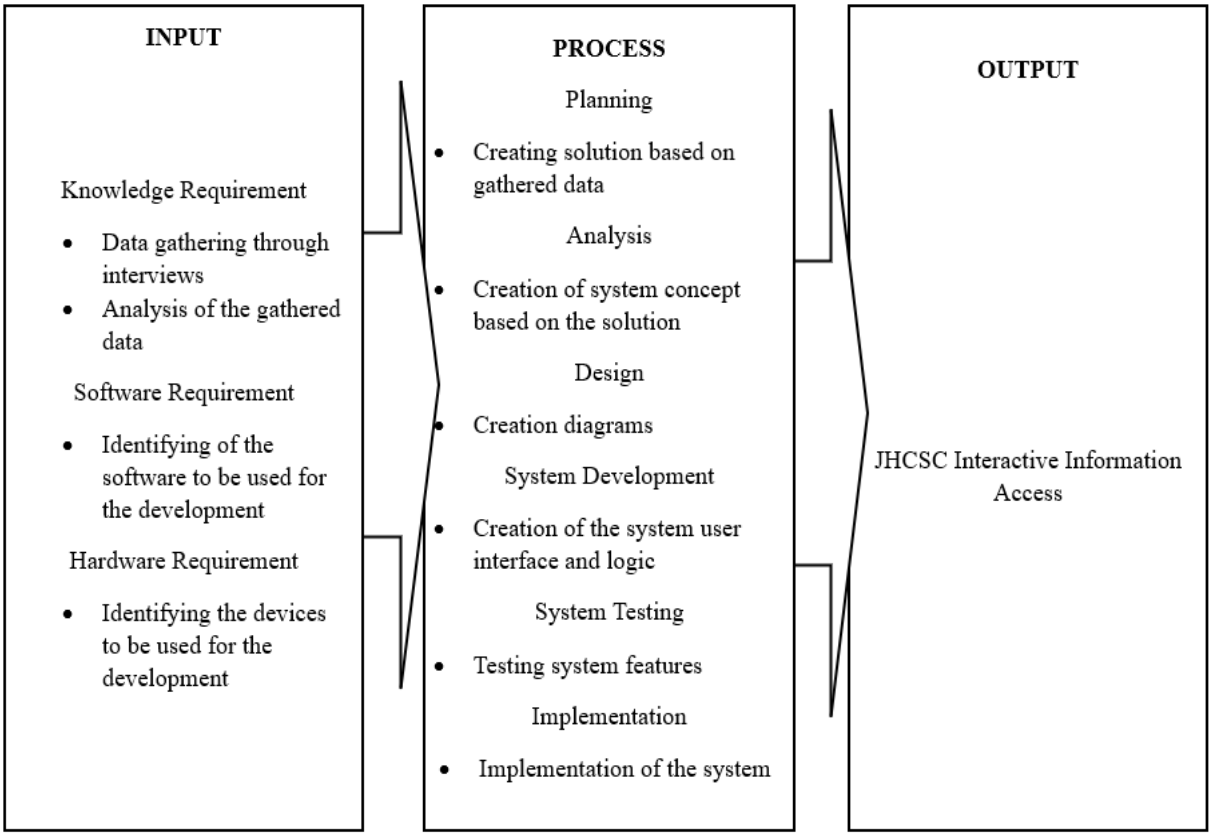



**Figure 4.** Conceptual Framework – IPO Diagram

The diagram above illustrates the study's conceptual framework using an Input-Process-Output (IPO) model. The input includes the knowledge requirement, which consists of data gathering through interviews and the analysis of the gathered data. The software requirements consist of identifying the software to be used for development, while the hardware requirements consist of identifying the devices to be used for the system's development. Next is the process stage of the IPO model. The process includes planning, which consists of creating a solution based on the gathered data. The analysis

involves developing the system concept based on the solution. The design includes creating diagrams that represent the entire system concept. System development involves creating the interface and the logic. System testing involves testing the system with its users. The implementation phase refers to the actual deployment of the system. The output stage will be the developed JHCSC Interactive Information Access.

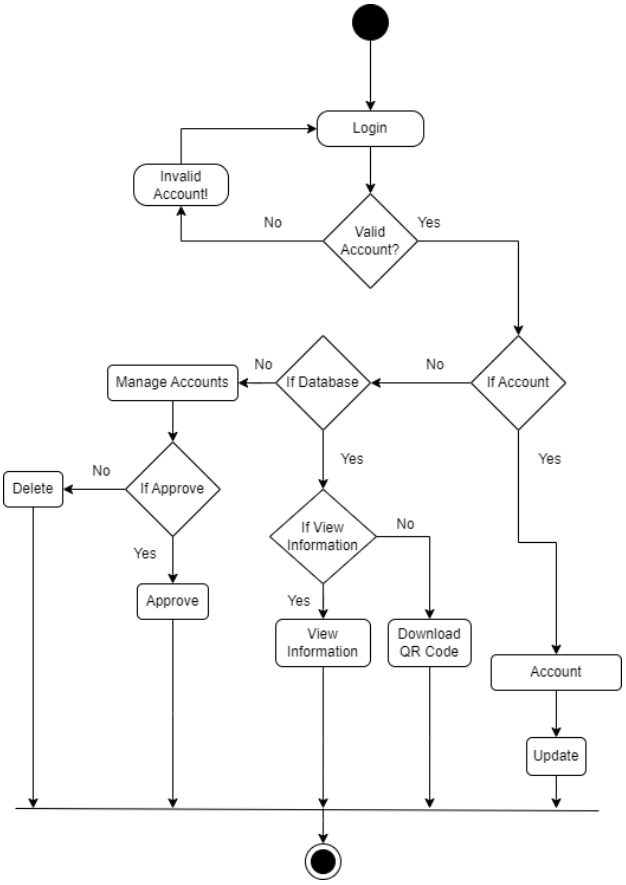



**Figure 5.** Activity Diagram (Administrator)

Figure 5 shows the activity diagram for the administrator. The admin must first log in to the system. After logging in, a condition will be encountered: "Valid account?" If not, an error message will be displayed and the user will be redirected to the login page. If the account is valid (yes), the admin can perform the following actions: the first is Manage Account, which allows the admin to approve or revoke users' access. Next is the Database, which allows the admin to view tree or building information or

download a QR code. Lastly, the Update Account activity allows the admin to update their own account information.

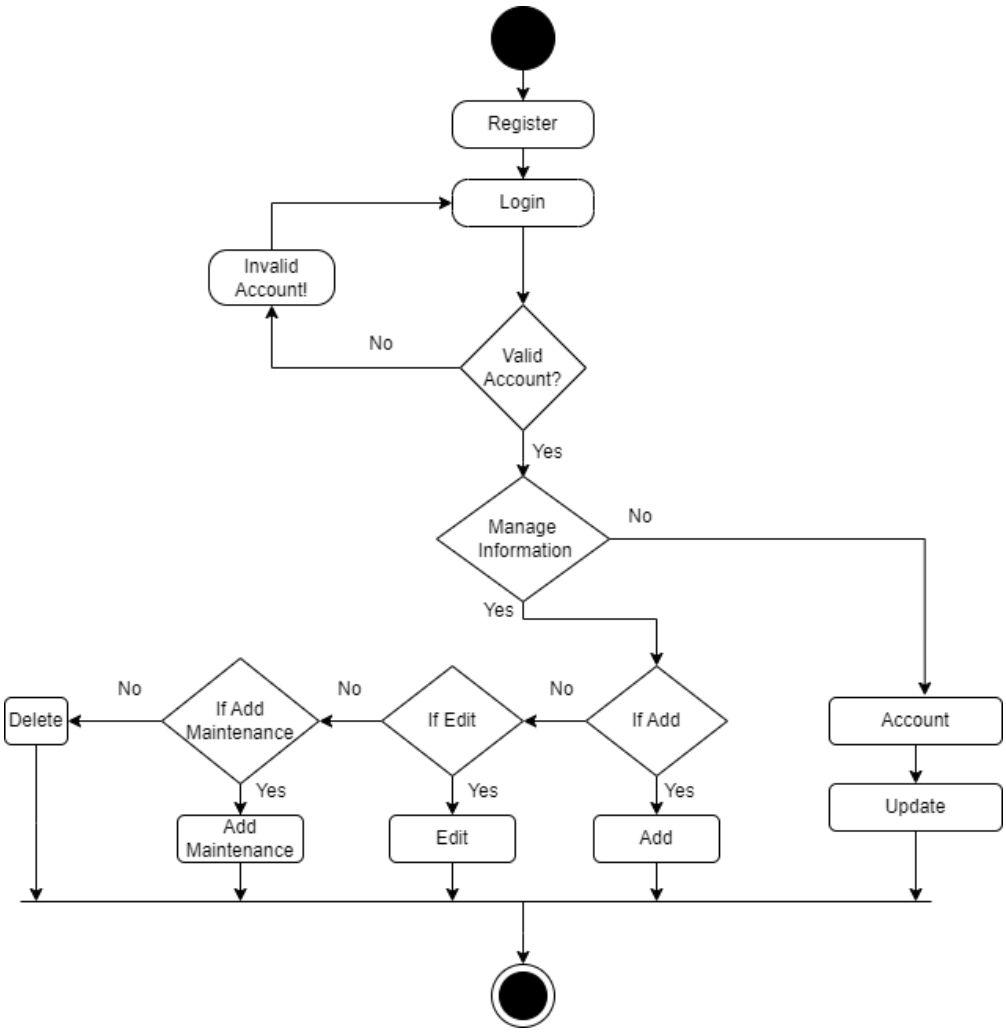



**Figure 6.** Activity Diagram (Instructor/Administrative Personnel)

Figure 6 shows the activity diagram for the instructor or administrative personnel. The user must first register in order to log in to the system. After logging in, a condition will be encountered: "Valid account?" If not, an error message will be displayed and the user will be redirected to the login page. If the account is valid (yes), the instructor will be able to perform the following actions: the first is Manage information, which allows users to add, edit, view, add maintenance, and delete

information. Another activity is the account that allow the users to modify their accounts.

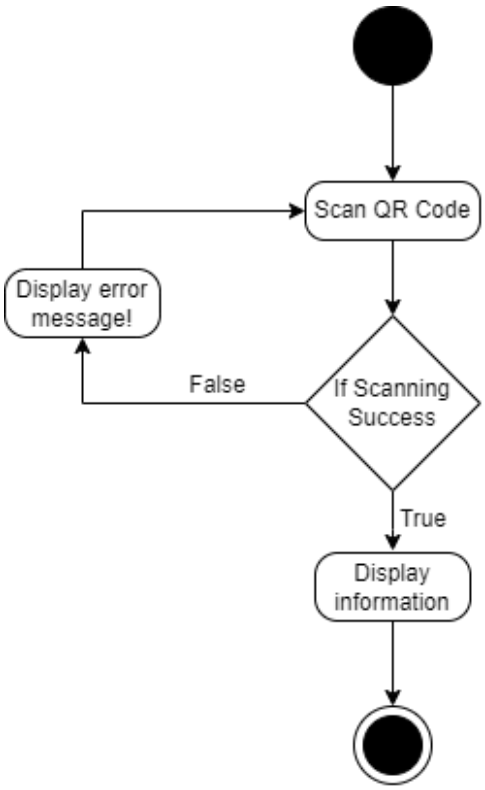



**Figure 7.** Activity Diagram (Public Users)

The diagram shown above illustrates the activity diagram for public users. Public users can scan QR codes attached to the trees and buildings. The system will check if the scanning is successful; if it fails, an error message will be displayed. If true, the information will be displayed.

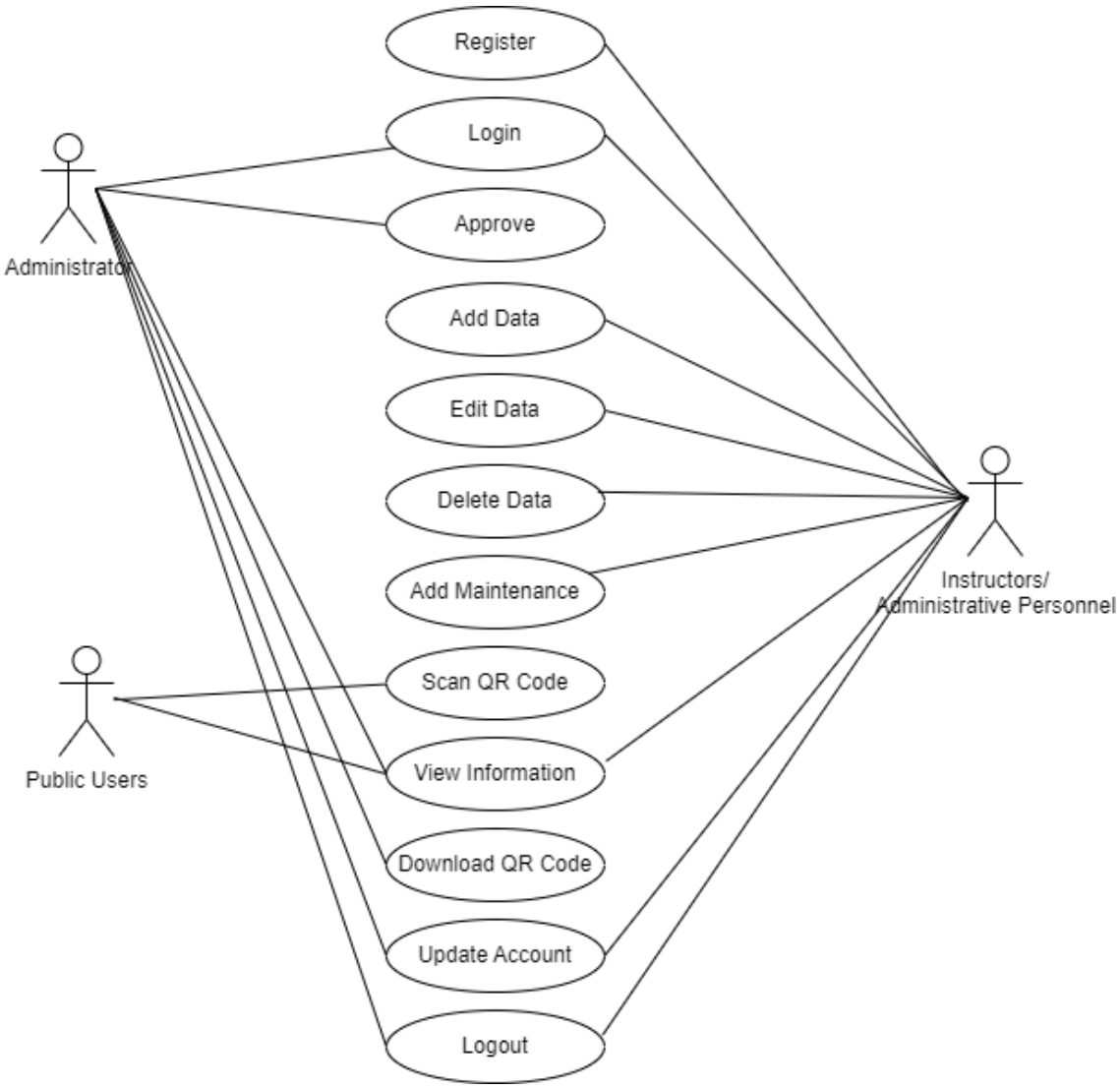



**Figure 8.** Use Case Diagram

The diagram shown above is the use case diagram. This diagram illustrates the different actions that users can perform within the system's available features. The first actor is the administrator, who can log in, approve accounts, view information, download QR codes, update their account, and log out. On the other hand, instructors or administrative personnel can register, log in, add data, edit tree data, delete data, add maintenance, view information, update their account, and log out. Public users can scan a QR code to view the corresponding information.

**Use Case Description**

| | |
|---|---|
| Name | Generate QR Code |
| Actors | Administrator |
| Description | The admin generates the QR code of the tree or buildings that contains its information. |
| Preconditions | The admin accesses the list of data saved in the system. |
| Postconditions | The QR code containing the tree information has been generated. |
| Main Course | 1. Open the tree or building information list.<br>2. Select a specific tree or building.<br>3. Click the generate QR Code button. |
| Alternate Course | The system encounters an error during the generation of the QR Code.<br>1. Display error message.<br>2. Back to Main Course Step 1. |

**Table 1.** Use Case Description for Generate QR Code

| | |
|---|---|
| Name | Scan QR Code |
| Actors | Public Users |
| Description | The public users scan the QR code to view the tree and building information. |
| Preconditions | The public users scan the QR code attached to the trees and buildings. |
| Postconditions | Tree information is displayed. |
| Main Course | 1. Scan the QR code attached to the trees and buildings.<br>2. The system checks the existence of data.<br>3. The system displays the information. |
| Alternate Course | The system encounters an error during the scanning of QR Code.<br>3. Display error message.<br>4. Back to Main Course Step 1. |

**Table 2.** Use Case Description for Scan QR Code

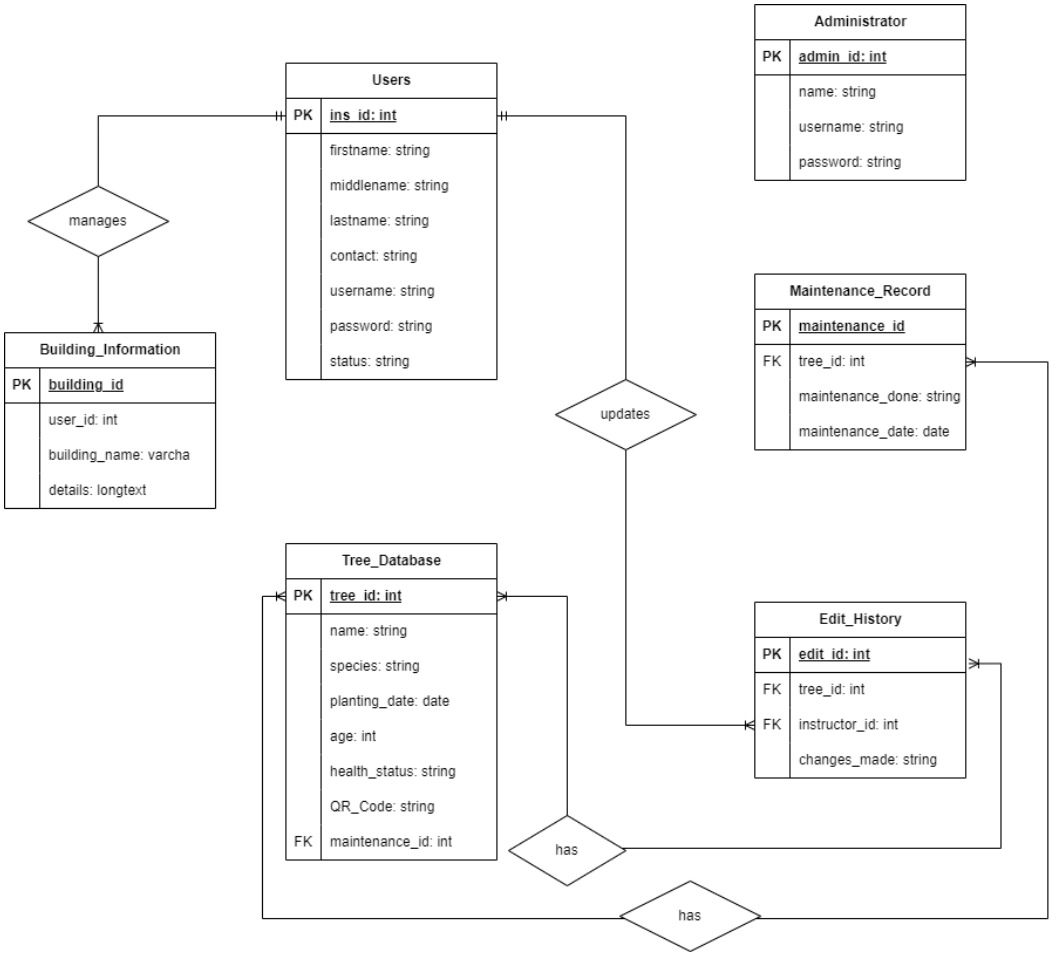



**Figure 9.** Entity Relationship Diagram

The diagram shown above is an Entity-Relationship Diagram. It illustrates that the entities Instructor and Edit History have a one-to-many relationship, as one instructor can update the information multiple times. The Tree Database and Edit History have a many-to-many relationship, since tree information can be updated multiple times. The Tree Database also has a many-to-many relationship with the Maintenance Record, as one or many trees can have one or many maintenance records. Also, the administrator table has no relationships with other entities, as it is a strong entity. Moreover, the Users and Building Information entity have a one-to-many relationship, since one user can manage one or many building information.

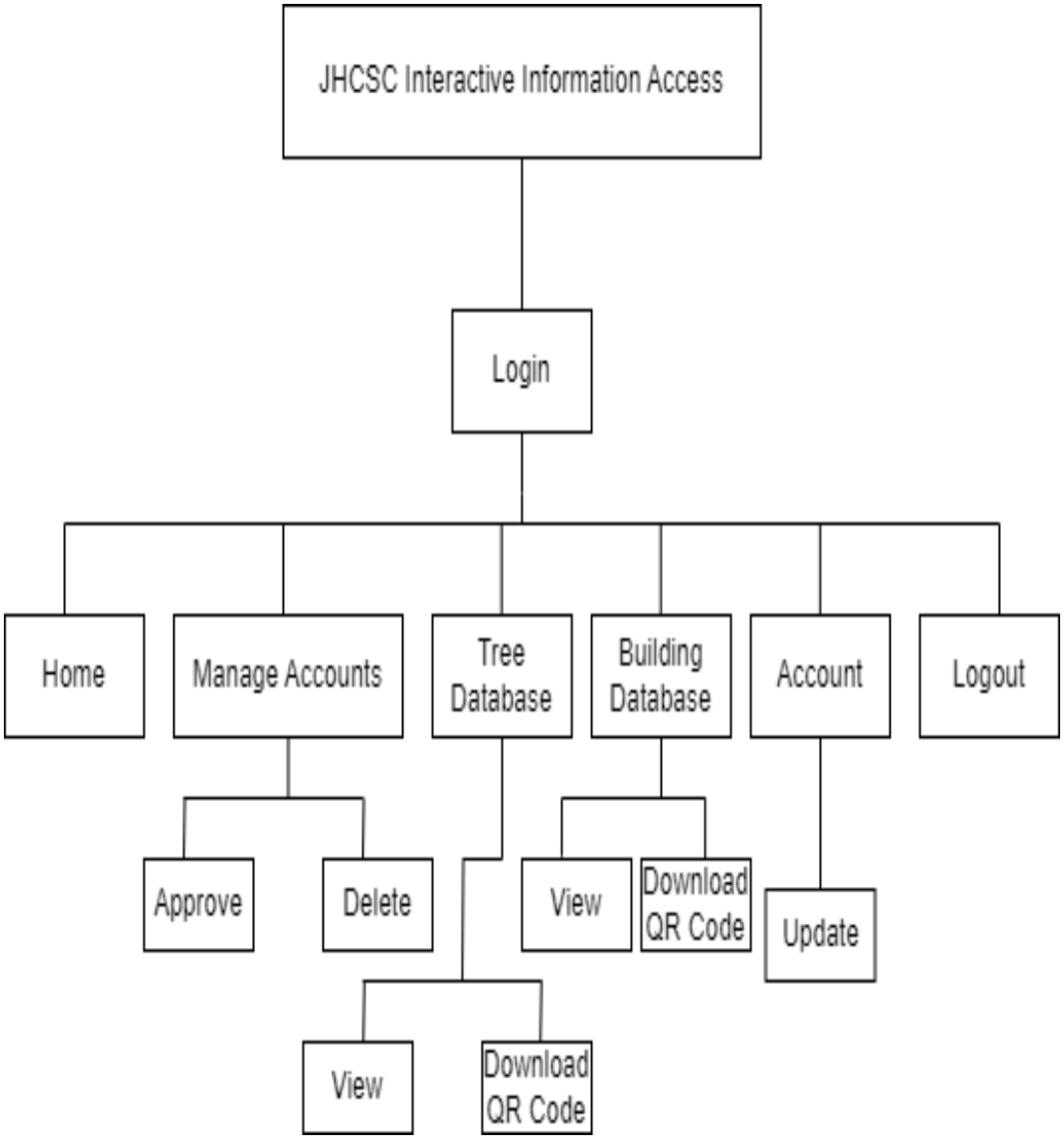



**Figure 10.** Structural Diagram (Administrator)

The diagram shown above is the structural diagram for the administrator that illustrates the structure of the system, including its menus and buttons. After logging into the system, the menus can be accessed. First is the home menu. Second is the manage accounts menu, which has approved and delete buttons. Third is the database menu, which has view and download QR code buttons. Fourth is the account menu, which has an update button. The last menu is logout.

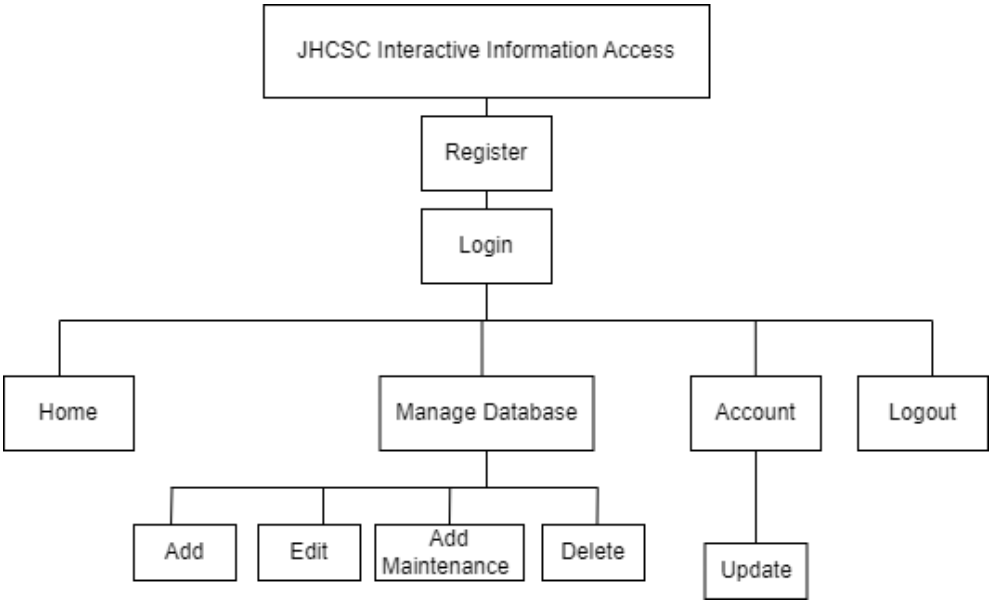



**Figure 11.** Structural Diagram for Instructor/Administrative Personnel

The diagram shown above is a structural diagram for instructors or administrative personnel that illustrates the system's structure, including its menus and buttons. After registering and logging into the system, the menus can be accessed. First is the home menu. Second is the manage information menu, which has added, edit, add maintenance, and delete buttons. Third is the account menu, which has an update button. The last menu is the logout.

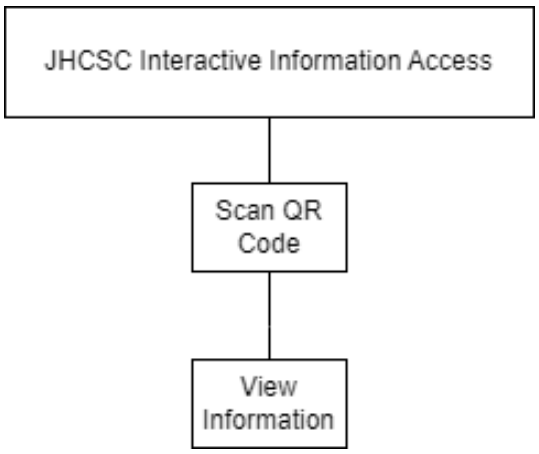



**Figure 12.** Structural Diagram (Public Users)

The diagram shown in Figure 12 is the structural diagram for public users that illustrates the system's structure, including its menus and buttons. After scanning the QR code, the users can view the information.

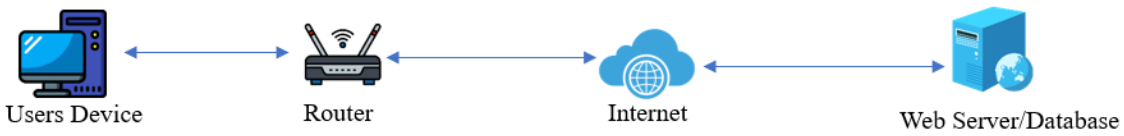



**Figure 13.** System Architecture

The diagram above shows the system architecture of the system. All user devices are connected to the router, which serves as the gateway for accessing the internet. Once connected to the internet, the devices can store and retrieve data from the web server and the database.

# CHAPTER IV

# DEVELOPMENT, TESTING, AND IMPLEMENTATION

This chapter discusses the development process of the JHCSC Interactive Information Access, the testing performed by the proponents, and the implementation plan for the system. It provides an overview of how the system was developed, its functionality, and the strategies used to ensure reliability and usability.

## Description of Prototype

The JHCSC Interactive Information Access is a web-based platform that provides easy, real-time access to information about both trees and school buildings on the J.H. Cerilles State College main campus. By integrating QR code technology with a centralized database, the system allows users to scan QR codes attached to trees or buildings and instantly view their corresponding profiles. For trees, the system displays information such as the common and local names, scientific name, family, ecological and conservation status, and maintenance history. For school buildings, QR codes provide details including the building name and its information.

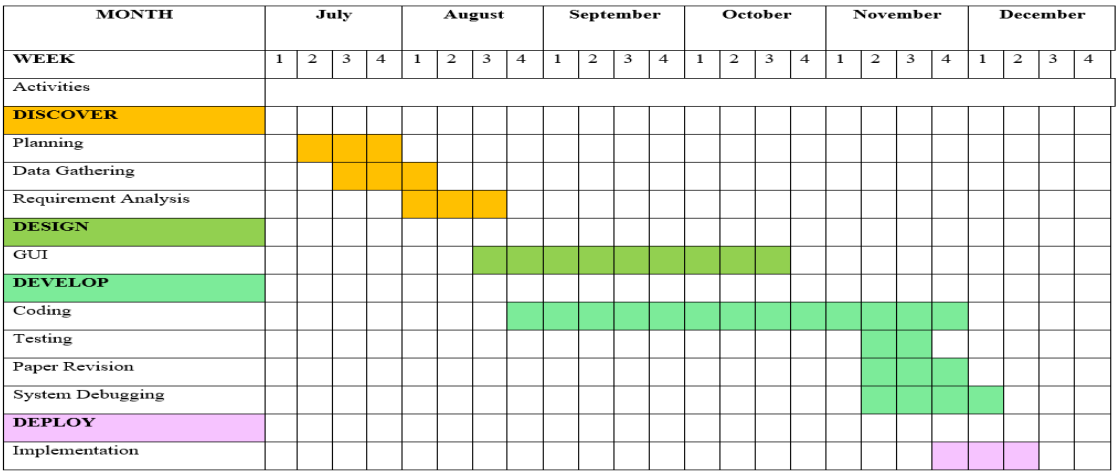



**Table 3.0 Gantt Chart**

The Gantt chart, as shown in Table 3.0, presents the project timeline from July to December for the development of the JHCSC Interactive Information Access. The Discover phase began with the planning stage during the second and third weeks of July, where the study's objectives, scope, and workflow were defined. This was followed by the data gathering stage, spanning from the third week of July to the second week of August, which involved collecting information from forestry instructors and administrative personnel to support system development. The requirement analysis phase, from the second to third week of August, focused on identifying the system requirements necessary for building the platform.

The Design phase took place from the fourth week of August to the third week of October, with a primary focus on creating a user-friendly, accessible graphical user interface (GUI). Following this, the development phase ran from the first week of September to the third week of November, during which the core functionalities of the system were coded and integrated, ensuring that QR code generation, database management, and real-time updates were implemented.

The Testing phase was carried out during the third and fourth weeks of November to evaluate the system's performance, functionality, and reliability. During the same period, the paper revision stage refined the documentation based on evaluators' feedback. Immediately afterward, the system debugging phase, from the fourth week of November to the first week of December, to resolve errors and enhance stability. Finally, the Deploy phase, from the fourth week of November to the fourth week of December, focused on implementing the system on the hosting server and conducting user orientation, and ensuring full operational support for all user roles.

**Development Phase**

The Development phase is the third stage of the project, where the proponents begin building the functionalities of the JHCSC Interactive Information Access system. Using PHP as the primary server-side scripting language, the different modules of the system were implemented to manage information about trees and school buildings. A MySQL database was created to securely store all relevant records, including tree profiles, building information, maintenance history, and user data. When a QR code is scanned, the system retrieves and displays the corresponding tree or building information in real time. Meanwhile, authorized personnel, such as instructors or administrative staff, can update or add new entries through the web-based interface, ensuring that all data remains current and accurate. The system's web-based platform was developed with HTML and CSS for structure and styling, making the interface user-friendly and responsive across devices. PHP handles dynamic processes such as generating unique QR codes for each tree and building, updating records, and controlling user access levels. XAMPP was used as the local development environment, allowing testing and debugging before deployment to the live server.

**Store Data Module**

The Store Data Module explains the lines of code that run in order to allow the user to store tree or building information in the database. The code *$common_name = trim($_POST['common_name']); $scientific_name = trim($_POST['scientific_name']); $family_name = trim($_POST['f amily_name']); $ecological_status = trim($_POST['ecological _status']); $conservation_status = trim($_POST['conservation_ status']); $brief_description = trim($_POST['brief_description']);* retrieves the submitted form data and removes

extra spaces. These values are necessary to create a complete profile for each tree, including its common name, scientific name, family, ecological and conservation status, and a brief description. The condition *if (!empty($common_name) && !empty($scientific_name))* checks whether the required fields are provided. If these fields are empty, the system executes *else { echo ' Common Name and Scientific Name are required! '; }* to notify that the data provided is invalid. The file upload section *if (isset($_FILES['closer_picture']) && $_FILES['closer_picture']['error'] == UPLOAD_ERR_OK) { ... } if (isset($_FILES['far_picture']) && $_FILES['far_picture']['error'] == UPLOAD_ERR_OK) { ... }* handles uploading the close and far images of the tree. It assigns a unique filename, moves the file to the *uploads/trees/* directory, and stores the path in variables *$closer_picture* and *$far_picture*. These images provide a visual reference for each tree. The prepared statement *$stmt = $pdo->prepare("INSERT INTO tree_database (common_name, scientific_name, family_name, ecological_status, brief_description, conversation_status, closer_picture, far_picture) VALUES (?, ?, ?, ?, ?, ?, ?, ?)"); $stmt->execute([...]);* inserts a new record into the tree_database table. Binding the variables ensures that the correct values are securely stored in the database.

*$common_name = trim($_POST['common_name']); $scientific_name = trim($_POST['scientific_name']); $family_name = trim($_POST['family_name']); $ecological_status = trim($_POST['ecological_status']); $conservation_status = trim($_POST['conservation_status']); $brief_description = trim($_POST['brief_description']); if (!empty($common_name) && !empty($scientific_name)) else { echo ' Common Name and Scientific Name are required! '; } if (isset($_FILES['closer_picture']) &&*

*$_FILES['closer_picture']['error'] == UPLOAD_ERR_OK) { ... } if (isset($_FILES['far_picture']) && $_FILES['far_picture']['error'] == UPLOAD_ERR_OK) { ... } $stmt = $pdo->prepare("INSERT INTO tree_database (common_name, scientific_name, family_name, ecological_status, brief_description, conversation_status, closer_picture, far_picture) VALUES (?, ?, ?, ?, ?, ?, ?, ?)"); $stmt->execute([...]); $tree_id = $pdo->lastInsertId();*

**Generate QR Code Module**

The Generate QR Code Module explains the lines of code that run in order to generate a QR code for a specific tree or building and store it in the database. The code *$qr_code = "https://api.qrserver.com/v1/create-qr-code/?size=180x180&data=JHCSC-Tree-ID:$tree_id";* creates a unique QR code URL using the newly inserted tree ID. This QR code will link to the tree's profile in the system. The prepared statement *$updateStmt = $pdo->prepare("UPDATE tree_database SET qr_code = ? WHERE tree_id = ?"); $updateStmt->execute([$qr_code, $tree_id]);* updates the tree_database table with the generated QR code URL for the corresponding tree. This ensures that each tree record now has a unique QR code that can be scanned to view its details.

*$qr_code = "https://api.qrserver.com/v1/create-qr-code/?size=180x180&data=JHCSC-Tree-ID:$tree_id"; $updateStmt = $pdo->prepare("UPDATE tree_database SET qr_code = ? WHERE tree_id = ?"); $updateStmt->execute([$qr_code, $tree_id]);*

**Scan QR Code Module**

The Scan QR Code Module explains the lines of code that run in order to allow a user to scan a QR code and display the corresponding tree details. The module uses

the Html5QrcodeScanner library to access the device camera and detect QR codes. The function *function onScanSuccess(qrMessage)* serves as the success callback when a QR code is detected. It retrieves the QR code content (qrMessage) and extracts the tree ID using the regular expression: *const match = qrMessage.match(/JHCSC-Tree-ID:(\d+)/);* If a valid tree ID is found, the system fetches the tree details from the server using *fetch(`get_tree_details.php?tree_id=${treeId}`) .then(response => response.json()) .then(data => { ... });* This retrieves the stored information for the scanned tree, including its common name, scientific name, family, ecological status, conservation status, images, and maintenance records. If the tree ID is invalid, an error message is shown: *showStatus('error', 'Invalid QR code format. Please scan a valid tree QR code.');* The function *function showTreeInfo(treeData, maintenanceRecords)* displays the fetched data in the HTML interface. It populates the tree details, adds images for close-up and far views, and renders maintenance history. The camera view is hidden while showing the tree info section for better user experience.

*<script> const statusMessage = document.getElementById('status-message'); const treeInfo = document.getElementById('tree-info'); const cameraContainer = document.getElementById('camera-container'); let html5QrcodeScanner = null; function onScanSuccess(qrMessage) { const match = qrMessage.match(/JHCSC-Tree-ID:(\d+)/); if (match && match[1]) { const treeId = match[1]; showStatus('info', 'Tree detected! Loading details...'); fetch(`get_tree_details.php?tree_id=${treeId}`) .then(response => response.json()) .then(data => { if (data.error) showStatus('error', data.error); else showTreeInfo(data.tree, data.maintenance); }) .catch(error => showStatus('error', 'Failed to load tree details')); } else { showStatus('error', 'Invalid QR code format.'); } if*

```
html5QrcodeScanner = new Html5QrcodeScanner("reader", { fps: isMobile ? 8 : 10, qrbox: qrboxSize, aspectRatio: 1.0, showTorchButtonIfSupported: true }); html5QrcodeScanner.render(onScanSuccess); showStatus('info', 'Camera started. Point at a QR code to scan.'); } function showTreeInfo(treeData, maintenanceRecords) { cameraContainer.style.display = 'none'; treeInfo.style.display = 'block'; document.getElementById('tree-name').textContent = treeData.common_name || 'Unknown'; document.getElementById('scientific-name').textContent = treeData.scientific_name || 'Unknown'; document.getElementById('family-name').textContent = treeData.family_name || 'N/A'; document.getElementById('ecological-status').textContent = treeData.ecological_status || 'N/A'; document.getElementById('tree-id').textContent = 'TR' + String(treeData.tree_id).padStart(3, '0'); // Add images and maintenance records similarly... } function showStatus(type, message) { const statusClasses = { success: 'status-success', error: 'status-error', info: 'status-info' }; statusMessage.innerHTML = `<div class="${statusClasses[type]}">${message}</div>`; } document.addEventListener('DOMContentLoaded', initScanner); </script>
```

**Alpha Testing**

Alpha Testing of the JHCSC Interactive Information Access System was conducted by the proponents to assess the system's core features, reliability, usability, efficiency, portability, and stability before its exposure to end users. This phase focused

on identifying technical issues, correcting errors, and ensuring that all system functions operated according to specifications.

**Functionality Test**

The purpose of this test is to evaluate whether all core system features function properly, including login authentication, data management (add, edit, delete), and QR code generation and scanning.

The system's main functional features were tested sequentially. Valid and invalid login credentials were entered to verify authentication. The tree and building information forms were completed and saved as test data. Existing records were edited and deleted to confirm database updates. QR codes were generated for selected records and scanned using a mobile device to verify accurate data retrieval.

The test passed. Users could log in with valid credentials, while invalid login attempts displayed appropriate error messages. Tree and building data were successfully added, updated, and deleted in the database. QR codes were generated correctly and, when scanned, displayed the correct corresponding information. (See Appendix A: Test Case #: 01)

**Usability Test**

The purpose of this test is to evaluate the clarity of the system interface, ease of navigation, visibility of controls, and overall user-friendliness.

Usability testing was conducted by allowing new users to explore the system without prior instructions to assess navigation clarity. Menus and buttons were examined to ensure visibility and readability. Invalid inputs were intentionally triggered to evaluate the clarity of error messages. Additionally, QR codes were scanned to test the ease and speed of information retrieval.

The test passed. New users were able to navigate the system without assistance. All menus and buttons were clearly visible and readable. Error messages were understandable and properly guided users when invalid inputs were entered. QR code scanning was smooth, and information loaded instantly. (See Appendix A: Test Case #: 02)

**Efficiency Test**

The purpose of this test is to measure the system's response time and overall performance during common user operations, including page loading, QR code processing, data retrieval, and simultaneous user access.

Efficiency testing was conducted by opening the tree and building information pages to measure load speed. QR codes were scanned to evaluate how quickly information was displayed. Large datasets were used to test database retrieval performance. Additionally, multiple users (10–20 concurrent users) were simulated to observe system responsiveness under moderate load conditions.

The test passed. The tree and building pages loaded within the expected 1–3 seconds. QR code information was displayed quickly after scanning. Large data lists were retrieved without noticeable delay. During simulations with 10–20 users, the system remained stable and responsive. (See Appendix A: Test Case #: 03)

**Portability Test**

The purpose of this test is to ensure that the system functions correctly across different web browsers, devices, screen resolutions, and operating systems.

Portability testing was conducted by accessing the system with different web browsers, including Chrome, Microsoft Edge, and Firefox, to verify cross-browser compatibility. The system was tested on both mobile and desktop devices to evaluate

display consistency. Various screen resolutions were used to check responsive layout behavior. Additionally, the system was tested on Windows and Android operating systems to assess stability and performance.

The test passed. The system functioned consistently across Chrome, Edge, and Firefox browsers. It displayed properly on both mobile and desktop devices with a responsive layout across different screen resolutions. The system also performed stably on Windows and Android platforms. (See Appendix A: Test Case #: 04)

**Perceived Usefulness Evaluation**

The purpose of this test is to determine how effective the JHCSC Interactive Information Access System is in helping users obtain meaningful and relevant information about trees and school buildings within the campus. The evaluation focused on how well the system supports quick, accurate, and efficient access to information through QR code technology.

Users scanned QR codes installed on selected trees and school buildings to access information. They also viewed detailed tree profiles containing scientific and local names. Observations were recorded during system interactions, and feedback was collected on the clarity, accessibility, and usefulness of the displayed information.

The test passed. Information from the three QR codes loaded instantly, and users reported that the tree details were easy to access. Scanning building QR codes displayed accurate and helpful information, particularly for campus visitors. Tree profiles clearly presented scientific and local names, which users found very informative. (See Appendix A: Test Case #: 05)

**Perceived Ease of Use Evaluation**

The purpose of this test is to assess how easily users can interact with the system without confusion or difficulty. This evaluation focused on the interface design, ease of navigation, QR code scanning process, and data management functions to determine whether tasks can be performed efficiently with minimal effort or technical knowledge.

Users were asked to navigate the system interface to observe layout clarity and menu organization. They scanned QR codes to test the simplicity and speed of the scanning process. Administrators performed data updates, including editing existing information, to evaluate the number of steps required and overall ease of use. Observations were recorded, and user feedback was collected during system interaction.

The test passed. Users reported that the system layout is simple and easy to understand. QR code scanning was completed without difficulty and functioned quickly. Administrative data updates required minimal steps, and users found editing information easy. (See Appendix A: Test Case #: 06)

**Load Testing**

Load testing was conducted to evaluate the performance, responsiveness, and stability of the JHCSC Interactive Information Access System under varying user traffic levels. The objective of this testing was to determine how well the system performs during normal, peak, and above-peak usage conditions, particularly during simultaneous QR code scanning and information retrieval. The table below presents the load testing processes, criteria, expected outcomes, actual results, and test status.

The load-testing table in Appendix A summarizes the load-testing results for the JHCSC Interactive Information Access System. The system demonstrated stable,

responsive performance under both normal and load conditions, particularly during simultaneous QR code scanning and information access. Database operations remained efficient even when multiple users accessed data concurrently.

Stress testing beyond the expected user load showed that the system remained stable without data loss and recovered smoothly after use. Additionally, mixed-operation testing confirmed that administrative updates did not affect public user access. Overall, the results indicate that the system can handle real-world usage demands on the JHCSC campus and is ready for deployment.

**Beta Testing**

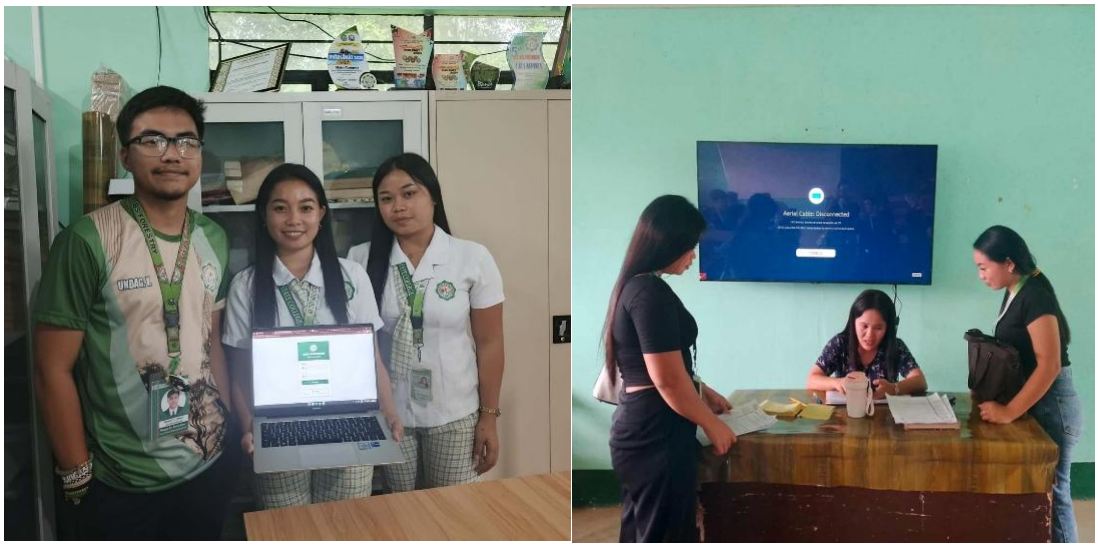


**Image 2.0 Beta Testing**

Beta testing was conducted to evaluate the user experience, satisfaction, and functionality of the JHCSC Interactive Information Access system in real-world conditions. The proponents allowed actual users to navigate and use the system independently.

As shown in the documentation above, Ms. Ivy M. Molina, a faculty member, together with the program chair's assistant, personally tested the system and expressed

satisfaction with its features, emphasizing that the platform is beneficial for viewing important information about trees and buildings in JHCSC.

| **Strategy** | **Activity** | **Persons Involve** | **Duration** |
|---|---|---|---|
| Uploading | Uploading the system online | Proponents | 1 Day |
| Training | Train users | Proponents & Users | 3 Weeks and 4 Days |

**Table 5.0 Implementation Plan**

The implementation plan presented in Table 5.0 shows that the system upload will take up to 1 day, including purchasing the domain and uploading the code. For training, it will be a consecutive 3-week, 4-day training, since many students cannot be trained in 1 session.

# CHAPTER V

# IMPLEMENTATION RESULT, DISCUSSION, CONCLUSION, AND RECOMMENDATION

This chapter illustrates the outcomes of the system's development, showing the user interface and presenting sample outputs. It further presents the study's conclusion, reflecting on the findings in light of the research objectives. The chapter also offers several recommendations, providing insights for future researchers and considerations for enhancing the study's context and environment.

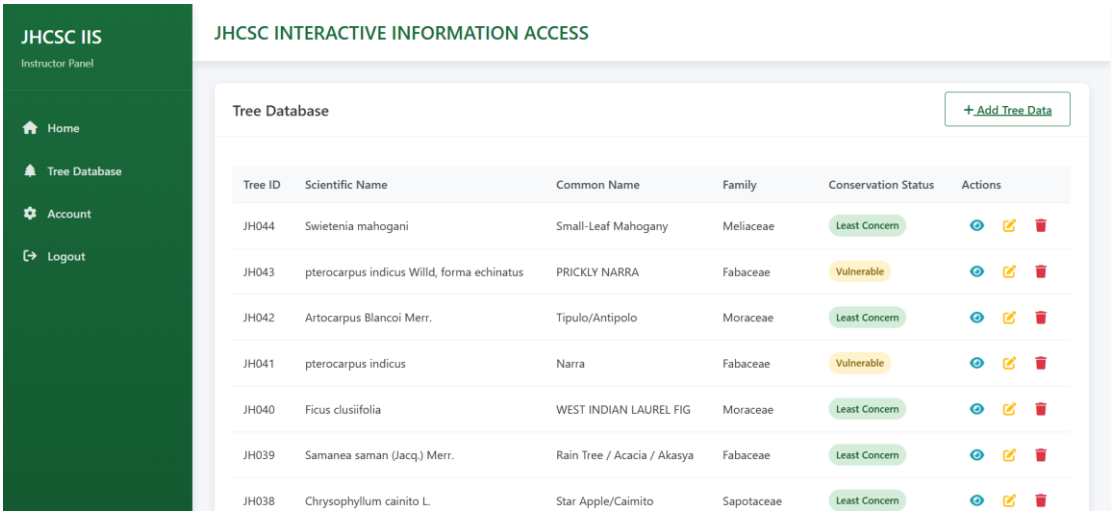



**Image 3.0 Manage Tree Data**

Image 3.0 shows the system interface when the instructor clicks on the tree database. The interface displays an "Add Tree Data" button, which the user can click to add new tree information. It also includes buttons for viewing, editing, and deleting, making it easy to manage tree data.

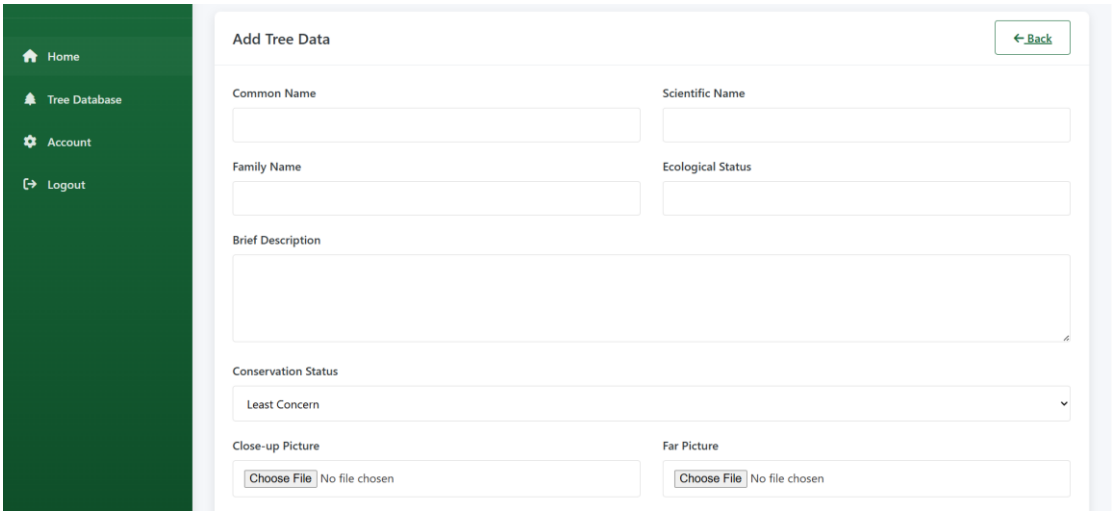



**Image 4.0 Add Tree Data**

Image 4.0 displays the user interface after the instructor clicks the "Add tree data" button. It is a form for the user's input, including details that need to be stored in the system.

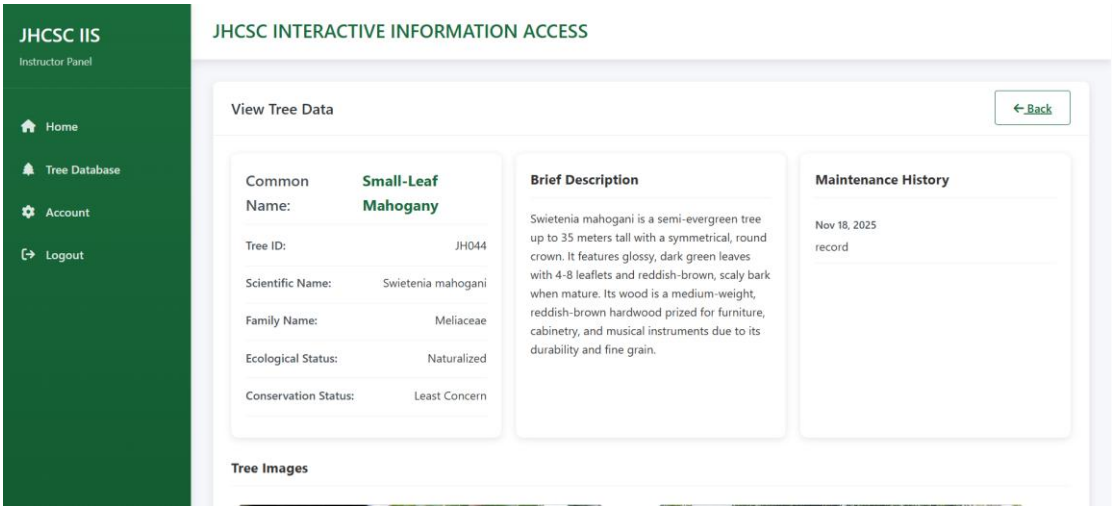



**Image 5.0 View Tree Data**

Image 5.0 displays the user interface when the view icon is clicked. This interface shows the details of the tree information stored in the database, allowing the instructor to easily view the data.

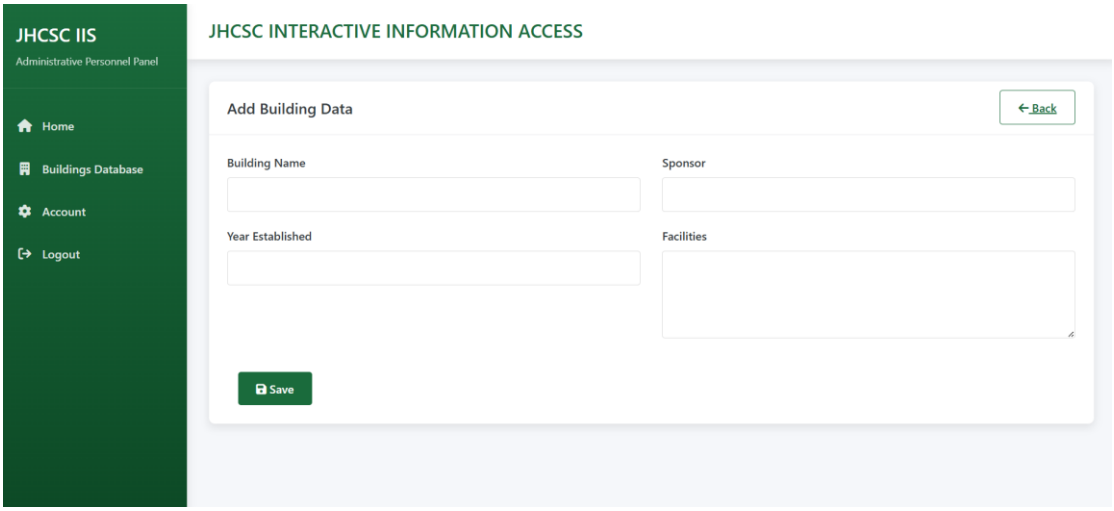



**Image 6.0 Add Building Data**

Image 6.0 shows the user interface after administrative personnel click the "Add tree data" button. It is a form for the user's input, including details that need to be stored in the system.

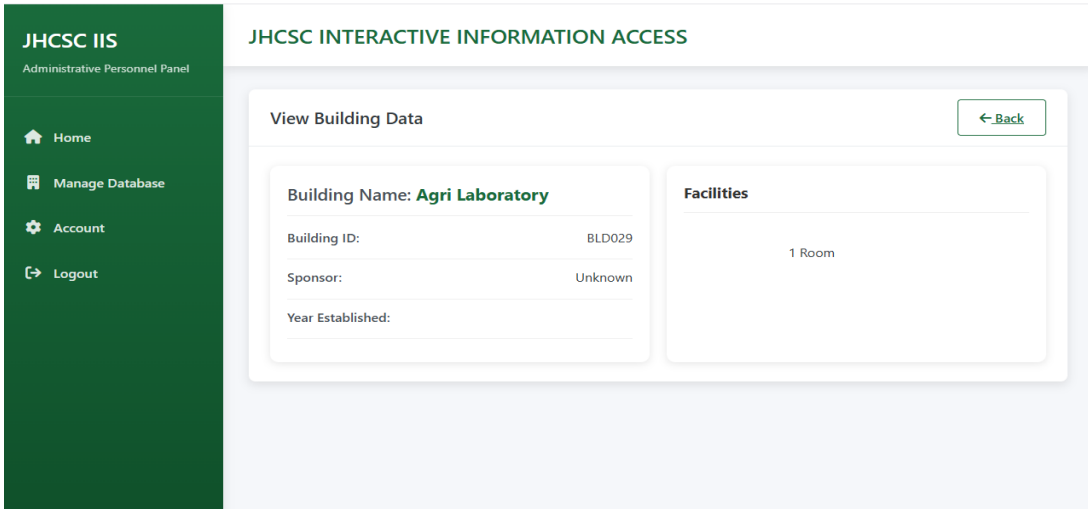



**Image 7.0 View Building Data**

Image 7.0 displays the user interface when the view icon is clicked. This interface shows the details of the building information stored in the database, allowing the instructor to easily view the data.

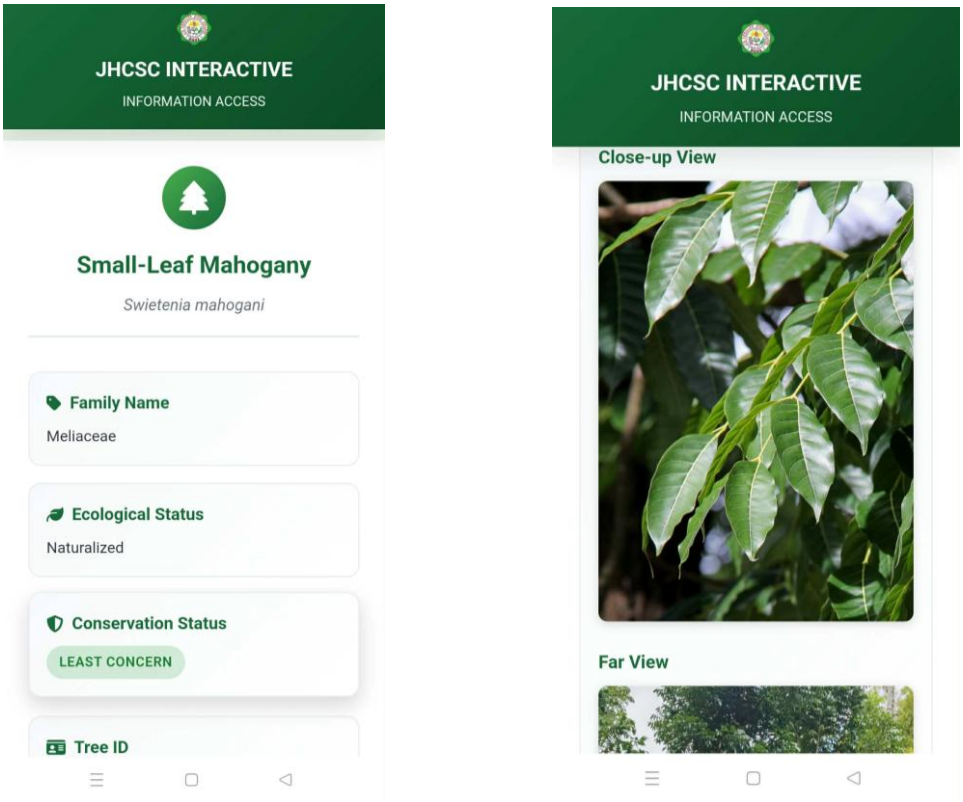



**Image 8.0 View Scanned QR Code**

Image 8.0 shows the user interface visible to public users after scanning the QR code. The interface allows public users to view the necessary data.

**Results and Discussions**

The results and discussion of the evaluation conducted on the JHCSC Interactive Information Access System. Data were gathered from 70 respondents using a 5-point Likert scale and analyzed using frequency distributions, weighted means, descriptive ratings, and rankings.

The interpretation of the weighted mean followed the scale: 4.21–5.00 (Strongly Agree), 3.41–4.20 (Agree), 2.61–3.40 (Neutral), 1.81–2.60 (Disagree), and 1.00–1.80 (Strongly Disagree). The summarized evaluation results are presented in Appendix A, the Survey Data Table, followed by a discussion of the findings.

The Survey Data table shows the respondents' results of the JHCSC Interactive Information Access System. The results show that all indicators obtained Agree (A) to Strongly Agree (SA) descriptive ratings, with an overall weighted mean of 4.32, indicating a very high level of user acceptance and satisfaction with the system.

The highest-ranked indicators were "Scanning the QR code and viewing information is fast" and "I can easily understand how to access and interpret the displayed information," both of which obtained a weighted mean of 4.37. These results indicate that the system is highly efficient and easy to understand, allowing users to access campus information quickly and effectively.

The statement "The system improves my access to accurate tree and building data" had the highest weighted mean of 4.43, ranking 1st, highlighting the system's strength in providing reliable, accurate campus information.

Usability-related indicators such as ease of learning, simplicity of navigation, and task completion efficiency also received Strongly Agree ratings, confirming that the system requires minimal effort to use and supports user tasks effectively.

The statement "The system user interface and element positioning are efficient" ranked last with a weighted mean of 4.20; however, it still received an Agree rating, indicating that the interface layout is generally acceptable and functional.

**Conclusion**

The development of the JHCSC Interactive Information Access successfully addressed the core problem of inefficient, outdated access to information about trees and buildings on the J.H. Cerilles State College main campus. By integrating QR code

technology with a web-based platform, the study successfully achieved its general and specific objectives. The system provides a digital repository for managing tree and building profiles, generates unique QR codes for real-time access, and provides a user-friendly interface for instructors, administrative personnel, and public users.

**Recommendations**

The following recommendations of this study are:

- Future researchers should consider expanding the system's scope to include other campus assets beyond trees and buildings, such as historical markers, artworks, or laboratory equipment.
- Adding features such as a searchable digital map of the campus showing the locations of all tagged trees and buildings would greatly enhance navigation.
- It is recommended that JHCSC institutionalize a long-term maintenance plan for the system.

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# APPENDICES

## APPENDIX A

### Test Case #: 01 – Functionality Test

| Test Case ID | Test Description | Steps | Expected Result | Actual Output |
|---|---|---|---|---|
| FUN-01 | Valid login | Enter correct username & password → Click Login | User successfully logs in to the dashboard | PASSED |
| FUN-02 | Invalid login | Enter incorrect credentials → Click Login | An error message is displayed | PASSED |
| FUN-03 | Add tree information | Fill out the tree form → Click Save | Tree data saved in the database | PASSED |
| FUN-04 | Add building information | Fill out building form → Click Save | Building data saved successfully | PASSED |
| FUN-05 | Generate QR code | Select tree/building → Click Generate QR | QR code generated and downloadable | PASSED |
| FUN-06 | Scan QR code | Scan QR using a mobile device | Correct tree/building info displayed | PASSED |
| FUN-07 | Edit information | Select record → Update details → Save | Updated data reflected in the system | PASSED |
| FUN-08 | Delete information | Select record → Click Delete | Record removed from the database | PASSED |

**Test Case #: 02 – Usability Test**

| Test Case ID | Test Description | Steps | Expected Result | Actual Output |
|---|---|---|---|---|
| USA-01 | Navigation clarity | New user explores menus | User navigates without assistance | PASSED |
| USA-02 | Button visibility | Check menus and buttons | All buttons are visible and readable | PASSED |
| USA-03 | Error message clarity | Trigger invalid input | A clear error message is shown | PASSED |
| USA-04 | QR scanning ease | Scan QR code | Information loads instantly | PASSED |

**Test Case #: 03 – Efficiency Test**

| Test Case ID | Test Description | Steps | Expected Result | Actual Output |
|---|---|---|---|---|
| EFF-01 | Page load speed | Open tree/building page | Loads within 1–3 seconds | PASSED |
| EFF-02 | QR display speed | Scan QR code | Info displayed quickly | PASSED |
| EFF-03 | Database retrieval | View large data list | No noticeable delay | PASSED |
| EFF-04 | Multiple users | Simulate 10–20 users | System remains responsive | PASSED |

**Test Case #: 04 – Portability Test**

| Test Case ID | Test Description | Steps | Expected Result | Actual Output |
|---|---|---|---|---|
| POR-01 | Cross-browser test | Use Chrome, Edge, Firefox | The system works consistently | PASSED |
| POR-02 | Device compatibility | Test on mobile & desktop | Proper display on all devices | PASSED |

| | | | | |
|---|---|---|---|---|
| POR-03 | Screen resolution | Test different resolutions | Responsive layout | PASSED |
| POR-04 | OS compatibility | Test Windows & Android | Stable performance | PASSED |

**Evaluation #: 05 – Perceived Usefulness Evaluation**

| **Test ID** | **Activity** | **Observation** | **User Feedback** |
|---|---|---|---|
| PU-01 | Scan the QR code on the tree | Information loads instantly | Tree details are easy to access. |
| PU-02 | Scan the QR code on the building | Accurate building info displayed | Helpful for visitors. |
| PU-03 | View tree profile | Clear scientific and local names | Very informative. |

**Test Case #: 06 – Perceived Ease of Use Evaluation**

| **Test ID** | **Activity** | **Observation** | **User Feedback** |
|---|---|---|---|
| PEU-01 | Navigate system interface | Simple layout | Easy to understand. |
| PEU-02 | Scan QR code | No difficulty encountered | Scanning is fast. |
| PEU-03 | Admin data update | Minimal steps required | Editing information is easy. |

**Load Testing Table**

| Purpose | Criteria / How to Test | Expected Outcome | Actual Output | Status |
|---|---|---|---|---|
| To evaluate system stability under normal operating load | • Simulate expected number of users scanning QR codes simultaneously• Test access to tree and building information | System responds correctly with acceptable loading time | Information loaded quickly and accurately with no noticeable delay | PASSED |
| To measure performance under peak user traffic | • Simulate multiple simultaneous QR scans (20–30 users)• Monitor response time and server behavior | System maintains responsiveness without performance degradation | System maintained fast response times during peak usage | PASSED |
| To test database performance during concurrent access | • Perform multiple simultaneous data retrieval requests• Monitor database response | Database processes requests without errors or delays | Data retrieval remained stable and error-free | PASSED |
| To evaluate system behavior beyond expected load | • Increase simultaneous users beyond expected peak• Observe system reaction under stress | System remains stable or degrades gracefully without data loss | System remained stable with no data loss and recovered after peak usage | PASSED |
| To test system stability during mixed operations | • Simultaneous QR scans while admin updates data• Monitor system consistency | System remains stable and reflects updates correctly | Updates were reflected instantly without affecting user access | PASSED |

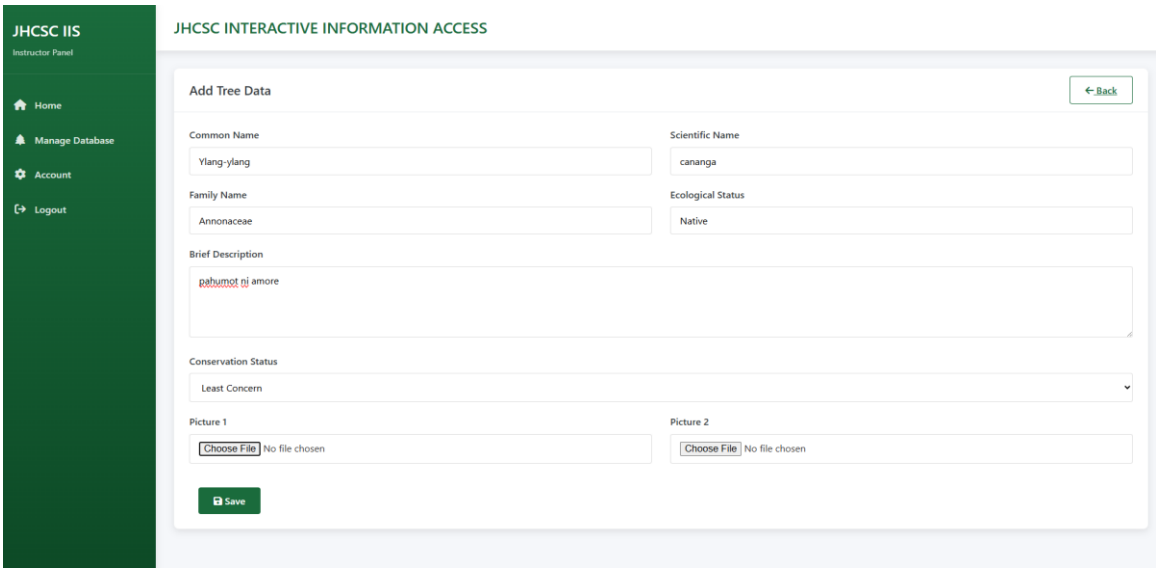
**Survey Data Table**

| Statement | 5 | 4 | 3 | 2 | 1 | TWP | WM | Descr. | Rank |
|---|---|---|---|---|---|---|---|---|---|
| The system user interface and element positioning are efficient. | 26 | 33 | 10 | 1 | 0 | 70 | 4.20 | A | 6 |
| I find the system interface simple and easy to navigate. | 30 | 30 | 10 | 0 | 0 | 70 | 4.29 | SA | 2 |
| Scanning the QR code and viewing information is fast. | 34 | 29 | 6 | 1 | 0 | 70 | 4.37 | SA | 1 |
| I can easily understand how to access and interpret the displayed information. | 32 | 32 | 6 | 0 | 0 | 70 | 4.37 | SA | 1 |
| Learning to use the Interactive Information Access System is easy for me. | 32 | 30 | 8 | 0 | 0 | 70 | 4.34 | SA | 3 |
| Using the Interactive Information Access System makes it easier to complete tasks related to campus information. | 27 | 36 | 6 | 1 | 0 | 70 | 4.27 | SA | 4 |
| The system improves my access to accurate tree and building data. | 36 | 28 | 6 | 0 | 0 | 70 | 4.43 | SA | 1 |

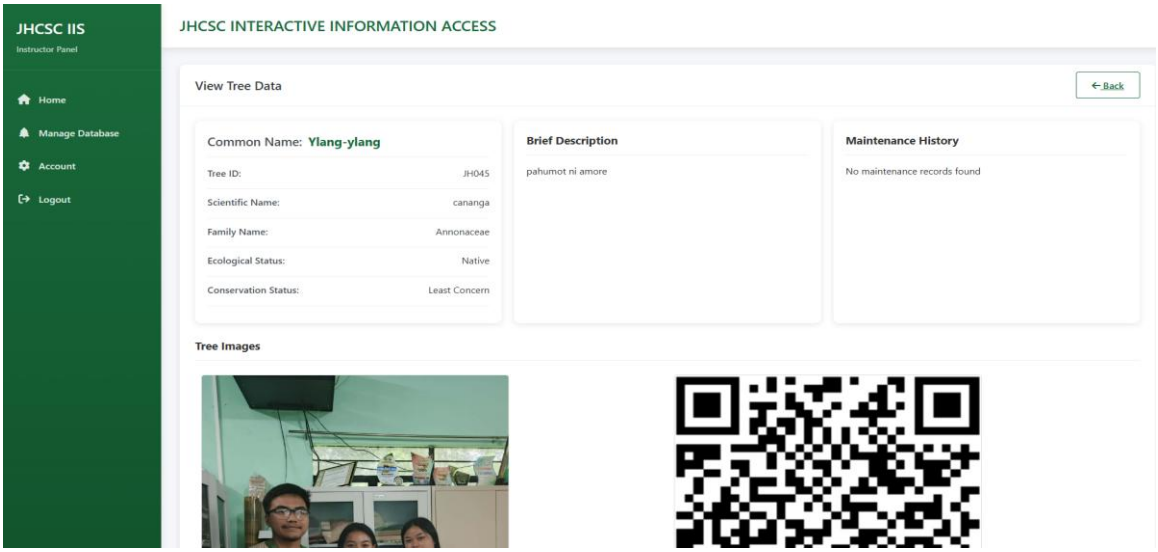
| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| The system enhances efficiency in locating campus information. | 29 | 33 | 8 | 0 | 0 | 70 | 4.30 | SA | 3 |
| The system does not display an error during its use. | 31 | 28 | 10 | 1 | 0 | 70 | 4.27 | SA | 4 |
| The overall functionality is very accurate. | 30 | 34 | 6 | 0 | 0 | 70 | 4.34 | SA | 3 |
| **TOTAL** | | | | | | **4.32** | **SA** | | |

# APPENDIX B

## SAMPLE INPUT AND OUTPUT



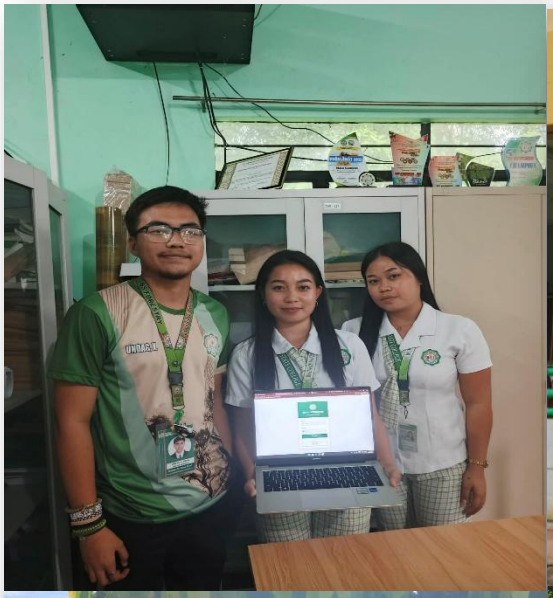

The image shown above is the sample input where users can input when using the system, which is stored in the system. This information will be saved and processed for filtering.




The image shown above is the sample output of this system, wherein the data during the inputs will be retrieved from the database and display it on the system output generation feature.

# APPENDIX C

## DOCUMENTATION















**APPENDIX D**

CURRICULUM VITAE



**Personal Data**

Name : Grecelyn Estribillo Aranjuez

Gender : Female

Date of Birth : March 08, 2003

Place of Birth : District Dapdap, Culabay, Tabina Zamboanga Del Sur

Civil Status : Single

Citizenship : Filipino

Religion : Roman Catholic

Address : District Dapdap, Culabay, Tabina Zamboanga Del Sur

Father's Name : Gregorio L. Aranjuez

Occupation : Farmer

Present Address : District Dapdap, Culabay, Tabina Zamboanga Del Sur

Mother's Name : Gundina E. Aranjuez

Occupation : Housekeeper

Present Address : District Dapdap, Culabay, Tabina Zamboanga Del Sur

**EDUCATIONAL BACKGROUND**

| **Education Graduated** | **School** | **Year** |
|---|---|---|
| Primary | : Culabay, Elemetary School | 2015-2016 |
| Junior High Scool | : Culabay National High School | 2019-2020 |
| Senior High School | : Culabay National High School | 2021-2022 |
| Tertiary | : JHCSC Main Campus | 2025-2026 |
| Degree | : Bachelor of Science in Information Technology | |



**APPENDIX D**

CURRICULUM VITAE

**Personal Data**

Name : Herlyn Dinopol Aranjuez

Gender : Female

Date of Birth : March 16, 2003

Place of Birth : District Manggostan, Culabay, Tabina Zamboanga Del Sur

Civil Status : Single

Citizenship : Filipino

Religion : Roman Catholic

Address : District Manggostan, Culabay, Tabina Zamboanga Del Sur

Father's Name : Hermey L. Aranjuez

Occupation : Farmer

Present Address : District Manggostan, Culabay, Tabina Zamboanga Del Sur

Mother's Name : Letecia D. Aranjuez

Occupation : Housekeeper

Present Address : District Manggostan, Culabay, Tabina Zamboanga Del Sur

**EDUCATIONAL BACKGROUND**

| **Education Graduated** | **School** | **Year** |
|---|---|---|
| Primary | : Culabay, Elemetary School | 2015-2016 |
| Junior High Scool | : Culabay National High School | 2019-2020 |
| Senior High School | : Culabay National High School | 2021-2022 |
| Tertiary | : JHCSC Main Campus | 2025-2026 |
| Degree | : Bachelor of Science in Information Technology | |

**APPENDIX D**

CURRICULUM VITAE



**Personal Data**

Name : Kissie Jay Buenavista

Gender : Female

Date of Birth : September 09, 1998

Place of Birth : District Quary, Kagawasan, Dimataling Zamboanga Del Sur

Civil Status : Single

Citizenship : Filipino

Religion : Roman Catholic

Address : District Quary, Kagawasan, Dimataling Zamboanga Del Sur

Father's Name : Loyd M. Buenavista

Occupation : Construction worker

Present Address : District Quary, Kagawasan, Dimataling Zamboanga Del Sur

Mother's Name : Teresita V. Buenavista

Occupation : Housekeeper

Present Address : District Quary, Kagawasan, Dimataling Zamboanga Del Sur

**EDUCATIONAL BACKGROUND**

| **Education Graduated** | **School** | **Year** |
|---|---|---|
| Primary | : San Miguel, Elemetary School | 2015-2016 |
| Junior High Scool 2020 | : Lureano Salusod National High School | 2019- |
| Senior High School 2022 | : Lureano Salusod National High School | 2021- |
| Tertiary | : JHCSC Main Campus | 2025-2026 |
| Degree | : Bachelor of Science in Information Technology | |

## APPENDIX D

CURRICULUM VITAE



**Personal Data**

Name : Angel L. Guinto

Gender : Female

Date of Birth : January 14, 2004

Place of Birth : Betinan, San Miguel Zamboanga Del Sur

Civil Status : Single

Citizenship : Filipino

Religion : INC

Address : Betinan, San Miguel Zamboanga Del Sur

Father's Name : Bertsman S. Guinto

Occupation : Deceased

Present Address : Betinan, San Miguel Zamboanga Del Sur

Mother's Name : Myrna L. Guinto

Occupation :Sales Lady

Present Address : Betinan, San Miguel Zamboanga Del Sur

**EDUCATIONAL BACKGROUND**

| **Education Graduated** | **School** | **Year** |
|---|---|---|
| Primary | : Betinan, Elemetary School | 2015-2016 |
| Junior High Scool | : Betinan, National High School | 2019-2020 |
| Senior High School | : Betinan, National High School | 2021-2022 |
| Tertiary | : JHCSC Main Campus | 2025-2026 |
| Degree | : Bachelor of Science in Information Technology | |